Sampling Edge Colorings at the Asymptotic Vizing Bound

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Abstract

We prove the rapid mixing of Wang–Swendsen–Kotecký dynamics on proper q-edge-colorings whenever $q \geq \Delta + \mathsf{C}\Delta^{13/14}$ for a positive universal constant C. This result asymptotically closes the gap between q-edge-colorability and the existence of an efficient sampling algorithm for proper q-edge colorings.

We establish a lower bound on the spectral gap using two comparison inequalities between the Dirichlet forms of WSK dynamics and single-edge Glauber dynamics in opposite directions. To handle the possible scarcity of locally available colors, we prove a weighted estimate that bounds the contribution of colorings with few available colors by an auxiliary Dirichlet form.

1 Introduction

Let $G = (V, E)$ be a finite simple graph with m edges and maximum degree Δ, and let $q \geq 1$ be an integer. A proper q-edge-coloring is a map $\sigma : E \to [q] \coloneqq \{1, \ldots, q\}$ that assigns different colors to edges sharing a vertex. Write Ω for the set of these colorings and μ for the uniform distribution on Ω. At least Δ colors are necessary for the proper edge coloring to exist, and Vizing's theorem [Viz64] guarantees that $\Delta + 1$ suffice.

Sampling proper colorings is a central problem in approximate counting and sampling. For vertex colorings of a graph of maximum degree Δ, a longstanding conjecture predicts polynomial-time approximate counting and sampling whenever $q \geq \Delta + 1$. The best known general bound is $q \geq 1.7408\Delta$ for sufficiently large Δ, proved in the recent preprint of Li and Oveis Gharan [LOG26, Theorem 1.7]. For edge colorings, Haqi [Haq26] recently proved rapid mixing with $2\Delta + 3$ colors, the best known bound.

The algorithms in [LOG26, Haq26] use the heat-bath version of Glauber dynamics: at each step, one edge selected uniformly at random is recolored by a uniformly random color from all available colors that keep the properness. However, this chain need not be irreducible even when substantially more than $\Delta + 1$ colors are available: some Δ-regular graphs have proper $(2\Delta - 1)$-edge-colorings from which no single edge can be recolored to a different color [DHP20].

1.1 Main results

In this work, we close the gap between the colorability and the existence of the sampling algorithm asymptotically by proving the rapid mixing of Wang–Swendsen–Kotecký (WSK) when $q \geq (1+o(1))\Delta$. Here $o(1)$ represents a function of Δ that vanishes when $\Delta \to \infty$.

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Definition 1.1 (WSK dynamics)**.** For $q \geq 2$, let $\mathcal{P} \coloneqq \binom{[q]}{2}$ be the set of unordered pairs of distinct colors. For $p \in \mathcal{P}$ and $\sigma \in \Omega$, define

$$E_p(\sigma) \coloneqq \{e \in E \mid \sigma(e) \in p\}, \qquad b_p(\sigma) \coloneqq \left\{ \omega \in \Omega \;\middle|\; \begin{array}{ll} \omega(e) \in p, & e \in E_p(\sigma), \\ \omega(e) = \sigma(e), & e \notin E_p(\sigma) \end{array} \right\}.$$

The WSK dynamics is the Markov chain on Ω that, from a current coloring σ, chooses $p \in \mathcal{P}$ uniformly at random and then chooses the next coloring uniformly from $b_p(\sigma)$.

Equivalently, after choosing a color pair p uniformly at random, the subgraph $(V, E_p(\sigma))$ consists of connected components of paths and even cycles colored by the color pair p alternatively. The colors of each path and cycle are then independently exchanged with probability $1/2$.

Definition 1.2 (Mixing time)**.** Let P be the WSK Markov operator on Ω. For $k \in \mathbb{Z}_{\geq 0}$ and $\sigma, \omega \in \Omega$, write $P^k(\sigma, \omega)$ for the (σ, ω) entry of the matrix of P^k, the probability of being at ω after k steps when starting from σ. For $\eta \in (0, 1)$, define the *mixing time of WSK dynamics* by

$$t_{\mathrm{mix}}(\eta) \coloneqq \inf \left\{ k \in \mathbb{Z}_{\geq 0} \;\middle|\; \max_{\sigma \in \Omega} \frac{1}{2} \sum_{\omega \in \Omega} \left| P^k(\sigma, \omega) - \mu(\omega) \right| \leq \eta \right\},$$

with $\inf \varnothing \coloneqq +\infty$.

We prove the following spectral-gap and mixing-time bounds as our main theorem.

Theorem 1.3 (Main theorem)**.** *There is a positive universal constant* C *such that, whenever*

$$q \geq \Delta + \mathsf{C}\Delta^{13/14},$$

the spectral gap of the WSK dynamics satisfies

$$\mathrm{gap}(P) = 1 - \lambda_2(P) > \binom{q}{2}^{-1}, \tag{1.1}$$

where $\lambda_2(P)$ *is the second-largest eigenvalue of its Markov operator. Moreover,*

$$t_{\mathrm{mix}}(\eta) \leq \left\lceil \binom{q}{2} \left[\frac{m}{2} \log q + \log \frac{1}{2\eta} \right] \right\rceil. \tag{1.2}$$

1.2 Proof sketch

1.2.1 Comparison with Glauber dynamics on certain eigenfunctions

Fix $t \in (0, 1/2]$ with $\Delta \leq (1-t)q$. Define the linear operators $S, T : \mathbb{R}^\Omega \to \mathbb{R}^\Omega$ by

$$\begin{aligned} S &\coloneqq \binom{q}{2}(\mathrm{Id} - P), \\ (Tg)(\sigma) &\coloneqq \frac{1}{2} \sum_{\substack{\omega \in \Omega \\ |\{e \in E : \omega(e) \neq \sigma(e)\}| = 1}} (g(\sigma) - g(\omega)), \end{aligned} \tag{1.3}$$

for $g \in \mathbb{R}^\Omega$ and $\sigma \in \Omega$. For $m \geq 1$, we note that $T = \frac{mq}{2}(\mathrm{Id} - P_{\mathrm{G}})$, where P_{G} is the Markov operator of the single-edge Glauber dynamics (Metropolis version)[1] defined in Section 2.3.

For a function $f : \Omega \to \mathbb{R}$, define the WSK and Glauber Dirichlet forms

$$\mathcal{E}_S(f) \coloneqq \mathbb{E}_{\sigma \sim \mu}[f(\sigma) \cdot (Sf)(\sigma)], \qquad \mathcal{E}_T(f) \coloneqq \mathbb{E}_{\sigma \sim \mu}[f(\sigma) \cdot (Tf)(\sigma)].$$

[1] Terminology follows Delcourt–Heinrich–Perarnau [DHP20]. Metropolis proposes uniformly from $[q]$, rejecting illegal recolorings while the heat-bath version samples uniformly among legal colors.

Theorem 3.1 (informal). *There is a constant $\mathsf{C}_1 > 0$ such that, for sufficiently large qt^{14}, every nonzero $f : \Omega \to \mathbb{R}$ with $Sf = \lambda f$ for some $\lambda \in (0, 1]$ satisfies*

$$\mathcal{E}_S(f) \le \mathsf{C}_1 t^{-8} \mathcal{E}_T(f).$$

Theorem 3.2 (informal). *There is a constant $\mathsf{C}_2 > 0$ such that, for sufficiently large qt^{14}, every nonzero $f : \Omega \to \mathbb{R}$ with $Sf = \lambda f$ for some $\lambda \in (0, 1]$ satisfies*

$$\mathcal{E}_T(f) \le \frac{\mathsf{C}_2}{qt^6} \mathcal{E}_S(f).$$

Combining the two gives

$$0 < \mathcal{E}_S(f) \le \frac{\mathsf{C}_1 \mathsf{C}_2}{qt^{14}} \mathcal{E}_S(f),$$

which excludes such an eigenfunction of S when $qt^{14} > \mathsf{C}_1 \mathsf{C}_2$. Since S is positive semidefinite, all its nonzero eigenvalues are greater than 1. Under the theorem's color bound, Kempe connectivity ensures that 1 is a simple eigenvalue of P. The spectral-gap assertion of Theorem 1.3 then follows from the definition $S = \binom{q}{2}(\mathrm{Id} - P)$.

For $\mathcal{E}_S(f) \le \mathsf{C}_1 t^{-8} \mathcal{E}_T(f)$, we simulate flips of short two-color paths by sequences of single-edge recolorings. For a long path or a cycle, we recolor one edge to a color outside the pair, cutting the component. The proof is in Section 5.

For $\mathcal{E}_T(f) \le \mathsf{C}_2 (qt^6)^{-1} \mathcal{E}_S(f)$, we start from two colorings that differ by one legal single-edge recoloring and couple their color-pair updates. The eigenfunction equation relates the initial difference of f to its differences after these updates. The proof is in Section 6.

Remark 1.4 (Choice of spectral cutoff). The spectral cutoff 1 in Theorems 3.1 and 3.2 is chosen for convenience. For every fixed $\lambda_\star > 0$, the same argument excludes eigenvalues of S in $(0, \lambda_\star)$ whenever

$$q \ge \Delta + \mathsf{C}_{\lambda_\star} \Delta^{13/14},$$

where $\mathsf{C}_{\lambda_\star} > 0$ depends only on $\lambda_\star$. Thus any fixed positive cutoff changes only the constant in the color bound, leaving the exponent 13/14 unchanged.

1.2.2 Controlling the few-available-color case by a Dirichlet form

Fix a color pair $p \in \mathcal{P}$ and an edge $e = \{u, v\} \in E$. For $\sigma \in \Omega$, let

$$H_{p,e}(\sigma) := \left|\{c \in [q] \setminus p \mid \sigma(f) \ne c \text{ for every } f \in E \text{ with } f \cap e \ne \varnothing\}\right|$$

be the number of colors except the color pair p that miss at both endpoints of e.

Theorem 4.1 (informal). *There are constants $\mathsf{A}, \mathsf{c} > 0$ such that, for large enough qt^3, every $g : \Omega \to \mathbb{R}$ supported on $\{\sigma \in \Omega \mid \sigma(e) \in p\}$ satisfies*

$$\mathbb{E}_\mu\left[g^2 \mathbf{1}_{\{H_{p,e} \le h_e\}}\right] \le \frac{\mathsf{A}}{q^2 t^6} \mathcal{E}_{\mathrm{aux},p}(g) + \mathsf{A} e^{-\mathsf{c} q t^3} \mathbb{E}_\mu[g^2],$$

where h_e is the threshold defined in (4.3) and $\mathcal{E}_{\mathrm{aux},p}$ is the Dirichlet form defined in (4.2).

This theorem asserts that when qt^3 is large enough, we cannot put arbitrarily large weights on bad cases where $H_{p,e}$ is small without magnifying a certain Dirichlet form of the weight function. It is used to avoid the bottleneck of the worst case where there are not enough colors to flip short paths or cut long paths and cycles in the first comparison. It is used to control the contribution of colorings with too few auxiliary colors to the Glauber Dirichlet form in the second comparison. The proof of Theorem 4.1 is in Section 4.

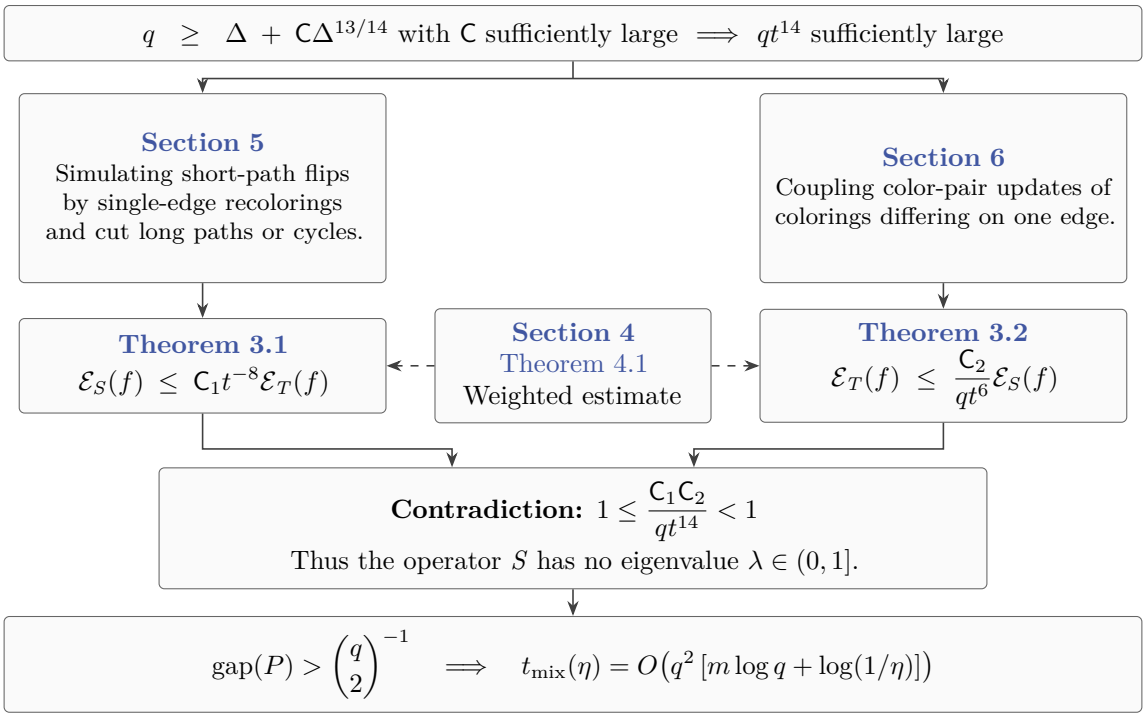


Figure 1: Proof outline of the main theorem. Dashed arrows show that Theorem 4.1 is used in the two comparison theorems.

1.3 Related works

Sampling colorings. For vertex colorings, the current best result is by Li and Oveis Gharan [LOG26]. They proved rapid mixing of Glauber dynamics for $q \geq 1.7408\Delta$ when Δ is sufficiently large, using local-to-global Bochner inequalities on cones. For edge colorings, this yields rapid mixing when $q \geq 3.4816(\Delta - 1)$ and Δ is sufficiently large.

The known bounds for edge colorings is better than this. Abdolazimi, Liu, and Oveis Gharan [ALOG21] used matrix trickle-down to prove rapid mixing of Glauber dynamics when $q \geq (10/3 + \varepsilon)\Delta$, for every fixed $\varepsilon > 0$ and sufficiently large Δ. Within the same framework, Wang, Zhang, and Zhang [WZZ24] improved the color bound to $q \geq 2\Delta + O(\Delta/\log \Delta)$ for sufficiently large Δ. Most recently, Haqi [Haq26] used the Bochner framework [CL26] to reduce it to $q \geq 2\Delta + 3$.

WSK dynamics. WSK dynamics was originally introduced for the antiferromagnetic Potts model in [WSK89, WSK90]. Earlier studies examined its numerical performance [FS99] and its ergodicity and nonergodicity on lattices [MS09, MS10, SS22]. For the mixing time anlaysis, Wang, Yoshida, and Zhang [WYZ26] developed colorwise entropy inequalities that yield rapid mixing on chordal and outerplanar graphs.

Comparison with Glauber dynamics. For such non-local chains, comparison with Glauber dynamics is a common approach to relating convergence rates. In the analyses of Vigoda [Vig00], Chen et al. [CDM+19], and Carlson and Vigoda [CV25], comparison of Dirichlet forms transfers rapid mixing from bounded-size flip dynamics to Glauber dynamics. Heinrich [Hei20] used a comparison with a single-component Kempe chain to prove rapid mixing of Glauber dynamics on chordal graphs. In the other direction, Łuczak and Vigoda [ŁV05] observed that rapid mixing of

Glauber dynamics implies rapid mixing of the single-component WSK variant studied in their paper.

1.4 Organization

Section 2 defines the notation, Markov operators, and Dirichlet forms. Section 3 states the two comparison theorems and uses them to prove Theorem 1.3. Section 4 proves the weighted missing-color estimate used in both comparisons. Section 5 bounds the WSK Dirichlet form by the Glauber Dirichlet form through short-path flips and cuts of long components. Section 6 proves the reverse bound using coupled color-pair updates. Appendix A constructs the component operators and proves the decomposition used in Section 5.

2 Preliminaries

2.1 Notation convention

Let $G = (V, E)$ have m edges and maximum degree Δ, let Ω be its set of proper q-edge-colorings, and let μ be uniform on Ω. Define the partition function $\mathsf{Z} := |\Omega|$. For a vertex $z \in V$, let $N(z) := \{w \in V \mid \{z, w\} \in E\}$ be its set of neighbors. For real functions f, g on Ω, write

$$\langle f, g\rangle_\mu := \mathbb{E}_\mu[fg], \qquad \|g\|_\mu^2 := \mathbb{E}_\mu[g^2].$$

For $\Xi \subseteq \Omega$ and $g : \Xi \to \mathbb{R}$, set

$$\|g\|_{\mu,\Xi}^2 := \mathsf{Z}^{-1} \sum_{\sigma\in\Xi} g(\sigma)^2.$$

For $g : \Omega \to \mathbb{R}$, the same notation applies to its restriction.

For self-adjoint linear operators A, B on real functions on a finite set X with positive measure ν, write $A \preceq B$, or equivalently $B \succeq A$, if

$$\langle f, Af\rangle_\nu \le \langle f, Bf\rangle_\nu \qquad \text{for every } f : X \to \mathbb{R}.$$

This is the Loewner order with respect to $\langle\cdot,\cdot\rangle_\nu$.

For a linear operator $A : \mathbb{R}^\Omega \to \mathbb{R}^\Omega$ that vanishes on constant functions and is self-adjoint and positive semidefinite with respect to $\langle\cdot,\cdot\rangle_\mu$, define its Dirichlet form by

$$\mathcal{E}_A(f) := \langle f, Af\rangle_\mu, \qquad f : \Omega \to \mathbb{R}.$$

2.2 WSK dynamics

Let $\mathcal{P} := \binom{[q]}{2}$ be the set of unordered pairs of distinct colors. We use p, r to denote color pairs, ranging over $\mathcal{P}$ unless otherwise specified. The operator P is the WSK Markov operator, and t_{mix} is its mixing time, as defined in Definition 1.2.

For $p \in \mathcal{P}$ and $\sigma \in \Omega$, recall that $E_p(\sigma) = \{e \in E \mid \sigma(e) \in p\}$ is the set of edges colored by p. The set $b_p(\sigma)$ consists of proper colorings that agree with σ outside $E_p(\sigma)$ and use only colors in p on $E_p(\sigma)$.

For $p = \{a, b\}$, let $s_p : [q] \to [q]$ exchange the colors a and b and act as the identity on all other colors. Let $K_p(\sigma)$ be the collection of edge sets of the paths and cycles formed by $E_p(\sigma)$, called *Kempe components*. For $A \subseteq E$, define $\Theta_{p,A} : [q]^E \to [q]^E$ by

$$(\Theta_{p,A}\sigma)(e) := \begin{cases} s_p(\sigma(e)), & e \in A, \\ \sigma(e), & e \notin A. \end{cases}$$

When $C \in K_p(\sigma)$, the map $\sigma \mapsto \Theta_{p,C}\sigma$ is a *component flip* and preserves the properness of the coloring.

WSK dynamics and its Dirichlet form. For a fixed p, if $\omega \in b_p(\sigma)$, then $b_p(\omega) = b_p(\sigma)$. Thus any two sets $b_p(\sigma)$ are equal or disjoint. So the sets $b_p(\sigma)$ form a partition of Ω. We define

$$K_p(\sigma,\omega) \coloneqq \frac{\mathbf{1}_{\{\omega\in b_p(\sigma)\}}}{|b_p(\sigma)|}, \qquad (P_p f)(\sigma) \coloneqq \frac{1}{|b_p(\sigma)|}\sum_{\omega\in b_p(\sigma)} f(\omega).$$

P_p is the orthogonal projection onto the functions constant on each set $b_p(\sigma)$ with respect to $\langle\cdot,\cdot\rangle_\mu$. Set $Q_p \coloneqq \mathrm{Id} - P_p$, the orthogonal projection onto the orthogonal complement of this subspace. For S from (1.3) and every $f : \Omega \to \mathbb{R}$, the projection property gives

$$\begin{aligned} P = \binom{q}{2}^{-1}\sum_{p\in\mathcal{P}} P_p, \qquad S = \sum_{p\in\mathcal{P}} Q_p, \\ \mathcal{E}_S(f) = \langle f, Sf\rangle_\mu = \sum_{p\in\mathcal{P}} \|Q_p f\|_\mu^2. \end{aligned} \tag{2.1}$$

2.3 Glauber dynamics and its Dirichlet form

For $e \in E$ and $c \in [q]$, write $\omega^{e\leftarrow c}$ for the coloring (not necessarily proper) obtained by changing $\omega(e)$ to c. The recoloring is *legal* if $c \neq \omega(e)$ and $\omega^{e\leftarrow c} \in \Omega$. For $e \in E$ and $p \in \mathcal{P}$, define $G_{e,p}$ by

$$(G_{e,p}f)(\omega) \coloneqq \begin{cases} \dfrac{f(\omega) - f(\omega^{e\leftarrow s_p(\omega(e))})}{2}, & \omega(e)\in p \text{ and } \omega^{e\leftarrow s_p(\omega(e))}\in\Omega, \\ 0, & \text{otherwise.} \end{cases}$$

With respect to $\langle\cdot,\cdot\rangle_\mu$, the operator $G_{e,p}$ is the orthogonal projection onto functions supported on colorings admitting a legal recoloring for e, p and taking opposite values on each such pair. Define the set of oriented legal recolorings, whose elements are also called *arcs*, by

$$A \coloneqq \{(\omega,\tau,e) \in \Omega\times\Omega\times E \mid \omega(e)\neq\tau(e),\ \omega(d)=\tau(d) \text{ for every } d\neq e\}.$$

Write $\omega \to \tau$ if $(\omega,\tau,e) \in A$ for some e. For $m \geq 1$, define the Markov operator P_{G} of the Glauber dynamics by

$$(P_{\mathrm{G}}g)(\omega) \coloneqq g(\omega) + \frac{1}{mq}\sum_{\substack{\tau\in\Omega\\ \omega\to\tau}} (g(\tau) - g(\omega)).$$

Recall that T, defined in (1.3), satisfies

$$T = \sum_{e\in E}\sum_{p\in\mathcal{P}} G_{e,p} = \frac{mq}{2}(\mathrm{Id} - P_{\mathrm{G}}).$$

Since each $G_{e,p}$ is an orthogonal projection, for every $f : \Omega \to \mathbb{R}$ the Dirichlet form of T satisfies

$$\mathcal{E}_T(f) \coloneqq \langle f, Tf\rangle_\mu = \sum_{e\in E}\sum_{p\in\mathcal{P}} \|G_{e,p}f\|_\mu^2 = \frac{1}{4}\sum_{(\omega,\tau,e)\in A} \mu(\omega)(f(\omega) - f(\tau))^2. \tag{2.2}$$

Both orientations of every legal recoloring belong to A.

Positive universal constants c, C may vary between inequalities.

2.4 Spectral gap and mixing time

For a reversible Markov operator K, define its *spectral gap* by

$$\mathrm{gap}(K) \coloneqq 1 - \lambda_2(K),$$

where $\lambda_2(K)$ is its second-largest eigenvalue.

Lemma 2.1 ([LP17, Theorem 12.4])**.** *Let K be the transition matrix of a finite, irreducible, aperiodic, reversible Markov chain on X with stationary distribution π. Define*

$$\pi_{\min} \coloneqq \min_{x\in X} \pi(x), \qquad \mathrm{gap}_*(K) \coloneqq 1 - \max\{|\lambda| \mid \lambda \neq 1 \text{ is an eigenvalue of } K\}.$$

For every $\eta \in (0,1)$, its total-variation mixing time, defined as in Definition 1.2, satisfies

$$t_{\mathrm{mix}}(K,\eta) \le \max\left\{0, \left\lceil \frac{1}{\mathrm{gap}_*(K)} \left[\frac{1}{2}\log\frac{1}{\pi_{\min}} + \log\frac{1}{2\eta}\right]\right\rceil\right\}.$$

If all eigenvalues of K are nonnegative, then $\mathrm{gap}_*(K) = \mathrm{gap}(K)$.

3 Proof of the main theorem

We derive Theorem 1.3 from the following two comparison theorems, proved in Sections 5 and 6, respectively.

Theorem 3.1. *There exist positive universal constants $\mathsf{C}_1, \mathsf{K}_1$ such that, for every $t \in (0, 1/2]$ with $\Delta \le (1-t)q$ and $qt^{14} \ge \mathsf{K}_1$, every nonzero function $f : \Omega \to \mathbb{R}$ with $Sf = \lambda f$ for some $\lambda \in (0,1]$ satisfies*

$$\mathcal{E}_S(f) \le \mathsf{C}_1 t^{-8} \mathcal{E}_T(f).$$

Theorem 3.2. *There exist positive universal constants $\mathsf{C}_2, \mathsf{K}_2$ such that, for every $t \in (0, 1/2]$ with $\Delta \le (1-t)q$ and $qt^{14} \ge \mathsf{K}_2$, every $f : \Omega \to \mathbb{R}$, and every $\lambda \in (0,1]$ satisfying $Sf = \lambda f$,*

$$\mathcal{E}_T(f) \le \frac{\mathsf{C}_2}{qt^6} \mathcal{E}_S(f).$$

Proof of Theorem 1.3. Let $\mathsf{C}_1, \mathsf{K}_1$ and $\mathsf{C}_2, \mathsf{K}_2$ be the constants in Theorems 3.1 and 3.2, respectively. Set

$$\mathsf{K}_* \coloneqq \max\{\mathsf{K}_1, \mathsf{K}_2, 2\mathsf{C}_1\mathsf{C}_2, 1\}, \qquad \mathsf{C} \coloneqq 2^{14}\mathsf{K}_*.$$

We may assume $m \ge 1$, so $\Delta \ge 1$. With $t \coloneqq \min\{(q-\Delta)/q, 1/2\}$, the hypothesis $q \ge \Delta + \mathsf{C}\Delta^{13/14}$ gives $qt^{14} \ge \mathsf{K}_*$. Indeed, if $q \le 2\Delta$, then

$$qt^{14} = \frac{(q-\Delta)^{14}}{q^{13}} \ge \frac{\mathsf{C}^{14}}{2^{13}} \ge \mathsf{K}_*,$$

if $q > 2\Delta$, then $qt^{14} = q/2^{14} \ge \mathsf{C}/2^{14} = \mathsf{K}_*$. Vizing's theorem [Viz64] ensures $\Omega \neq \varnothing$. Since $q \ge \Delta + 3$, the connectivity theorem in [Moh06, Theorem 3.1] implies that any two colorings in Ω are connected by a sequence of two-color component flips. This connectivity ensures that 1 is a simple eigenvalue of P. If a scalar eigenvalue $\lambda \in (0,1]$ existed with a nonzero eigenfunction $f : \Omega \to \mathbb{R}$, then Theorems 3.1 and 3.2 would give

$$0 < \mathcal{E}_S(f) \le \mathsf{C}_1 t^{-8} \mathcal{E}_T(f) \le \frac{\mathsf{C}_1\mathsf{C}_2}{qt^{14}} \mathcal{E}_S(f) \le \frac{1}{2}\mathcal{E}_S(f),$$

a contradiction. Thus S has no eigenvalues in $(0,1]$. Since S is positive semidefinite, all its nonzero eigenvalues exceed 1.

By definition (1.3), $P = \mathrm{Id} - \binom{q}{2}^{-1} S$, so

$$\mathrm{gap}(P) = 1 - \lambda_2(P) > \binom{q}{2}^{-1},$$

This establishes the spectral-gap assertion of Theorem 1.3. Applying Lemma 2.1 with $\pi_{\min} = \mathsf{Z}^{-1} \geq q^{-m}$ gives the required mixing-time bound. This proves Theorem 1.3. □

4 A weighted estimate for missing colors

In this section, we prove Theorem 4.1, an estimate that controls the weighted contribution from colorings in which an edge has few legal recolorings to colors outside a specified pair.

Throughout this section, we fix $t \in (0, 1/2]$ with $\Delta \leq (1-t)q$, a color pair $p \in \binom{[q]}{2}$, and an edge $e = \{u, v\} \in E$. Define the set-valued map $M_{p,e} : \Omega \to 2^{[q]\setminus p}$ and the function $H_{p,e} : \Omega \to \mathbb{Z}_{\geq 0}$ by

$$\begin{aligned} M_{p,e}(\sigma) &\coloneqq \{c \in [q] \setminus p \mid \sigma(f) \neq c \text{ for every } f \in E \text{ with } f \cap e \neq \varnothing\}, \\ H_{p,e}(\sigma) &\coloneqq |M_{p,e}(\sigma)|. \end{aligned} \tag{4.1}$$

Thus $M_{p,e}(\sigma)$ is the set of colors outside p missing at both endpoints of e, and $H_{p,e}(\sigma)$ is its cardinality. Write $H \coloneqq H_{p,e}$ throughout this section. Note that H is an integer-valued function on Ω.

Define the subset $D_{p,e} \coloneqq \{\sigma \in \Omega \mid \sigma(e) \in p\}$ of Ω. Define the quadratic form $\mathcal{E}_{\mathrm{aux},p} : \mathbb{R}^\Omega \to [0, \infty)$ by

$$\mathcal{E}_{\mathrm{aux},p}(g) \coloneqq \sum_{\substack{r \in \binom{[q]}{2} \\ r \cap p = \varnothing}} \|Q_r g\|_\mu^2, \qquad g \in \mathbb{R}^\Omega. \tag{4.2}$$

This is the Dirichlet form we use to control the weighted contribution from colorings with too few colors outside p missing at both endpoints of e. Define the positive constants κ, d_e, h_e, and h by

$$\begin{aligned} \kappa \coloneqq 10^{-3}, \qquad d_e &\coloneqq \max\{q - \deg(u), q - \deg(v)\}, \\ h_e \coloneqq \kappa t^2 d_e, \qquad h &\coloneqq \kappa t^3 q. \end{aligned} \tag{4.3}$$

Choose the endpoint labels so that $q - \deg(u) = d_e$. κ is chosen arbitrarily. The quantity d_e is the larger of the numbers of colors missing at u and v, and h_e is the threshold for too few colors missing at both endpoints used in Theorem 4.1. Since $d_e \geq tq$, the threshold h is a lower bound for h_e.

Theorem 4.1. *There are positive universal constants $\mathsf{A}, \mathsf{c}, \mathsf{K}_a$ such that, for every $t \in (0, 1/2]$, every integer q with $\Delta \leq (1-t)q$ and $qt^3 \geq \mathsf{K}_a$, every color pair $p \in \binom{[q]}{2}$, every edge $e = \{u, v\} \in E$, and every function $g : \Omega \to \mathbb{R}$ supported on $D_{p,e}$, the following bound holds:*

$$\mathbb{E}_\mu[g^2 \mathbf{1}_{\{H \leq h_e\}}] \leq \frac{\mathsf{A}}{q^2 t^6} \mathcal{E}_{\mathrm{aux},p}(g) + \mathsf{A} e^{-\mathsf{c} q t^3} \|g\|_\mu^2. \tag{4.4}$$

Here $\{H \leq h_e\} \coloneqq \{\sigma \in \Omega \mid H(\sigma) \leq h_e\}$. The inequality also holds with h in place of h_e.

We introduce notation for the color sets used at the endpoints, their differences, and their intersection. For each $z \in \{u, v\}$, define

$$A_z(\sigma) \coloneqq \{\sigma(f) \mid f \in E \text{ is incident to } z\} \setminus p.$$

Define

$$B_u(\sigma) \coloneqq A_u(\sigma) \setminus A_v(\sigma), \qquad B_v(\sigma) \coloneqq A_v(\sigma) \setminus A_u(\sigma), \qquad A_{uv}(\sigma) \coloneqq A_u(\sigma) \cap A_v(\sigma).$$

Define $U, V : \Omega \to \mathbb{Z}_{\geq 0}$ by

$$U(\sigma) \coloneqq \big|\{(\beta, \gamma) \in B_u(\sigma) \times B_v(\sigma) \mid u, v \text{ lie in distinct components of } (V, E_{\{\beta,\gamma\}}(\sigma))\}\big|,$$
$$V(\sigma) \coloneqq \big|\{(\beta, \gamma) \in A_{uv}(\sigma) \times M_{p,e}(\sigma) \mid u, v \text{ lie in distinct components of } (V, E_{\{\beta,\gamma\}}(\sigma))\}\big|.$$

The values $U(\sigma)$ and $V(\sigma)$ denote the numbers of pairs of distinct Kempe components containing u and v, indexed by their color pair outside p. For $U(\sigma)$, one color is used at u but not v and the other at v but not u. For $V(\sigma)$, one color is used at both endpoints and the other is missing at both.

Remark 4.2. For $\sigma \in D_{p,e}$, the component pairs represented by $U(\sigma)$ and $V(\sigma)$ determine the possible changes of H under a WSK update of a color pair r disjoint from p. The value of H can change only when u and v have degree one in the two-color subgraph and lie in distinct Kempe components. If the colors on their incident edges are different, as in the definition of $U(\sigma)$, then H increases by one with probability $1/2$. If those colors are the same, as in the definition of $V(\sigma)$, then H decreases by one with probability $1/2$. Otherwise, H remains unchanged.

4.1 An auxiliary bound

Lemma 4.3. *There exist positive universal constants $\mathsf{C}, \mathsf{K}_{\mathrm{d}}$ such that, whenever $qt^3 > \mathsf{K}_{\mathrm{d}}$, for every $g : \Omega \to \mathbb{R}$ supported on $D_{p,e}$ and every function $\theta \coloneqq \vartheta \circ H : \Omega \to [0, 1]$, with $\vartheta : \{0, \ldots, q\} \to [0, 1]$ and θ supported on $\{H \leq 2h_e\}$, we have*

$$\mathbb{E}_\mu[\theta(U - 2V)g^2] \geq \frac{t^2 q d_e}{10} \mathbb{E}_\mu[\theta g^2] - \frac{\mathsf{C}q}{t^2 d_e} \mathcal{E}_{\mathrm{aux},p}(g). \tag{4.5}$$

We prove this result through pointwise estimates and conditional expectations over two-color components.

For each $\beta \in [q] \setminus p$, let $z_\beta(\sigma)$ be the unique neighbor z of u with $\sigma(\{u, z\}) = \beta$ when $\beta \in B_u(\sigma)$, and leave it undefined otherwise. Define $S_\beta : \Omega \to 2^V$ by

$$S_\beta(\sigma) \coloneqq \begin{cases} \{z \in N(z_\beta(\sigma)) \mid \sigma(\{z_\beta(\sigma), z\}) \in B_v(\sigma)\}, & \beta \in B_u(\sigma), \\ \varnothing, & \beta \notin B_u(\sigma). \end{cases}$$

For each $c \in [q] \setminus p$, we define

$$(K_c(S_\beta))(\sigma) \coloneqq \big|\{z \in S_\beta(\sigma) \mid \sigma(f) = c \text{ for some edge } f \text{ incident to } z\}\big|.$$

Lemma 4.4. *For every $\sigma \in D_{p,e}$ and every $\beta \notin p$,*

$$\begin{aligned} U(\sigma) &\geq |B_u(\sigma)|(|B_v(\sigma)| - 1) - \sum_{\beta \in B_u(\sigma)} (K_\beta(S_\beta))(\sigma), \\ |S_\beta(\sigma)| &\leq |B_v(\sigma)| \leq \Delta, \qquad V(\sigma) \leq H(\sigma)\Delta. \end{aligned} \tag{4.6}$$

In particular, $\sum_{\gamma \notin p} (K_\gamma(S_\beta))(\sigma) \leq \Delta |S_\beta(\sigma)|$.

Proof. Fix $\sigma \in D_{p,e}$ and $\beta \in B_u(\sigma)$. We bound $U(\sigma)$ from below using color pairs whose Kempe component incident to u has one or two edges and does not contain v. If $\gamma \in B_v(\sigma)$ is missing at $z_\beta(\sigma)$, the $\{\beta,\gamma\}$-component containing u consists of the single edge $\{u, z_\beta(\sigma)\}$. If $\sigma(\{z_\beta(\sigma), z\}) = \gamma \in B_v(\sigma)$, this component consists of $\{u, z_\beta(\sigma)\}$ and $\{z_\beta(\sigma), z\}$ whenever $z \neq v$ and β is missing at z. Each case contributes to $U(\sigma)$. The resulting counts give the first inequality. The second bound is immediate: the edges from $z_\beta(\sigma)$ to vertices of $S_\beta(\sigma)$ have distinct colors in $B_v(\sigma)$, and $|B_v(\sigma)| \le \deg(v) \le \Delta$. The final bound follows directly from the degree bound, since each vertex of $S_\beta(\sigma)$ has at most Δ incident colors. The bound on V follows from $|A_{uv}(\sigma)| \le \Delta$ and $|M_{p,e}(\sigma)| = H(\sigma)$. □

Fix distinct $\beta, \gamma \in [q] \setminus p$ and put $r \coloneqq \{\beta,\gamma\} \in \binom{[q]}{2}$. For $\sigma \in \Omega$, let $K_r^\circ(\sigma)$ be the set of nonempty Kempe components of $(V, E_r(\sigma))$ that contain no edge incident to u or v. Define $k(\sigma) \coloneqq |K_r^\circ(\sigma)|$. We denote the members of $K_r^\circ(\sigma)$ by $C_1, \dots, C_{k(\sigma)}$. For $I \subseteq \{1, \dots, k(\sigma)\}$, define the coloring $\sigma_r^I : E \to [q]$ by

$$\sigma_r^I(f) \coloneqq \begin{cases} \gamma, & f \in \bigcup_{i\in I} C_i \text{ and } \sigma(f) = \beta, \\ \beta, & f \in \bigcup_{i\in I} C_i \text{ and } \sigma(f) = \gamma, \\ \sigma(f), & f \notin \bigcup_{i\in I} C_i. \end{cases}$$

The coloring σ_r^I is obtained by flipping the Kempe components C_i with $i \in I$. Flipping Kempe components preserves properness, so $\sigma_r^I \in \Omega$. Define the Markov operator Π_r by

$$(\Pi_r g)(\sigma) \coloneqq 2^{-k(\sigma)} \sum_{I \subseteq \{1,\dots,k(\sigma)\}} g(\sigma_r^I), \qquad g \in \mathbb{R}^\Omega, \quad \sigma \in \Omega.$$

It is the Markov operator of the transition that flips each r-Kempe component disjoint from the endpoints of e independently with probability $1/2$.

Remark 4.5. The operators Π_r and P_r both average over Kempe component flips. The difference is that P_r allows all r-Kempe components to flip, while Π_r leaves those meeting u or v fixed.

Lemma 4.6. *For every pair of distinct colors $\beta, \gamma \notin p$, with $r \coloneqq \{\beta,\gamma\}$, the operator Π_r is an orthogonal projection and*

$$0 \preceq \mathrm{Id} - \Pi_r \preceq Q_r.$$

Proof. The distinct sets $\{\sigma_r^I \mid I \subseteq \{1, \dots, k(\sigma)\}\}$, as σ ranges over Ω, form a partition of Ω. The operator Π_r averages uniformly over each set, so it is the orthogonal projection onto the functions constant on these sets with respect to $\langle \cdot, \cdot \rangle_\mu$. Each set is contained in $b_r(\sigma)$, so every function constant on the sets $b_r(\sigma)$ is fixed by Π_r. Consequently, $\Pi_r P_r = P_r \Pi_r = P_r$. Therefore

$$\|g - P_r g\|_\mu^2 = \|g - \Pi_r g\|_\mu^2 + \|\Pi_r g - P_r g\|_\mu^2, \qquad g \in \mathbb{R}^\Omega.$$

□

Define $D_{\beta\gamma} : \Omega \to \mathbb{R}$ by

$$D_{\beta\gamma}(\sigma) \coloneqq \big(K_\beta(S_\beta)\big)(\sigma) - \big(K_\gamma(S_\beta)\big)(\sigma),$$

and write $D \coloneqq D_{\beta\gamma}$ when the pair of indices is fixed.

Lemma 4.7. *Fix distinct colors $\beta, \gamma \notin p$, put $r := \{\beta, \gamma\}$, and set $D := D_{\beta\gamma}$. For every $\sigma \in \Omega$ and every $I \subseteq \{1, \dots, k(\sigma)\}$, the following quantities remain unchanged under the Kempe flips indexed by I:*

$$A_u(\sigma_r^I) = A_u(\sigma), \quad A_v(\sigma_r^I) = A_v(\sigma), \quad H(\sigma_r^I) = H(\sigma), \quad S_\beta(\sigma_r^I) = S_\beta(\sigma).$$

For every $\sigma \in \Omega$,

$$(\Pi_r D)(\sigma) \le 4, \qquad |D(\sigma)| \le \Delta. \tag{4.7}$$

Proof. The first three invariance identities hold because only components disjoint from u, v are flipped. For the last equality, if $\beta \in B_u(\sigma)$ then $z_\beta(\sigma_r^I) = z_\beta(\sigma)$: all r-edges at this vertex belong to the component containing $\{u, z_\beta(\sigma)\}$, which is excluded from $K_r^\circ(\sigma)$. If $\beta \notin B_u(\sigma)$, both sets S_β are empty.

Only endpoints of r-components can contribute to $D(\sigma)$. For every component in $K_r^\circ(\sigma)$, the average of its contribution over its two orientations is zero, since $S_\beta(\sigma_r^I) = S_\beta(\sigma)$. At most two components meet u or v, and they have at most four endpoints in total. This gives the first bound. The second follows from $|S_\beta(\sigma)| \le \Delta$. If $\beta \notin B_u(\sigma)$, then $D(\sigma) = 0$. □

Given a function $g : \Omega \to \mathbb{R}$ and a function $\vartheta : \{0, \dots, q\} \to [0, 1]$, define $\theta : \Omega \to [0, 1]$ by $\theta(\sigma) := \vartheta(H(\sigma))$. For every $\beta \notin p$, define $w_\beta : \Omega \to [0, 1]$ by

$$w_\beta(\sigma) := \theta(\sigma)\mathbf{1}_{\{\beta \in B_u(\sigma)\}}.$$

We call the functions w_β the *expectation weights*. Define the nonnegative scalars $\mathsf{R}, \mathsf{R}_\beta, \mathsf{E}_{\beta\gamma}$, writing $r = \{\beta, \gamma\}$ in the last expression, by

$$\mathsf{R} := \mathbb{E}_\mu[\theta g^2], \qquad \mathsf{R}_\beta := \mathbb{E}_\mu[w_\beta g^2], \qquad \mathsf{E}_{\beta\gamma} := \mathbb{E}_\mu[w_\beta((\mathrm{Id} - \Pi_r)g)^2].$$

Lemma 4.8. *For every $g : \Omega \to \mathbb{R}$, every function $\theta := \vartheta \circ H : \Omega \to [0, 1]$, and every ordered pair of distinct colors $\beta, \gamma \notin p$, with $r := \{\beta, \gamma\}$, the quantities defined above satisfy*

$$\mathbb{E}_\mu[w_\beta D_{\beta\gamma} g^2] \le (4 + \kappa t^2 d_e)\mathsf{R}_\beta + \frac{1001q^2}{t^2 d_e}\mathsf{E}_{\beta\gamma}.$$

Proof. Write $D := D_{\beta\gamma}$. Lemma 4.7 supplies the invariance identities and the pointwise bounds used below. The identities above imply $w_\beta(\sigma_r^I) = w_\beta(\sigma)$. Consequently, for every $F \in \mathbb{R}^\Omega$,

$$\Pi_r(w_\beta F) = w_\beta \Pi_r F.$$

Define the functions $a_{\beta\gamma}, b_{\beta\gamma} : \Omega \to \mathbb{R}$ by $a_{\beta\gamma} := \Pi_r g$ and $b_{\beta\gamma} := (\mathrm{Id} - \Pi_r)g$. Conditional expectation and orthogonality give

$$\begin{aligned} \mathbb{E}_\mu[w_\beta a_{\beta\gamma} b_{\beta\gamma}] &= 0, \\ \mathbb{E}_\mu[w_\beta a_{\beta\gamma}^2] &= \mathsf{R}_\beta - \mathsf{E}_{\beta\gamma} \le \mathsf{R}_\beta, \\ \mathbb{E}_\mu[w_\beta D a_{\beta\gamma}^2] &= \mathbb{E}_\mu[w_\beta a_{\beta\gamma}^2 \Pi_r D] \le 4\mathsf{R}_\beta. \end{aligned}$$

Expanding $g = a_{\beta\gamma} + b_{\beta\gamma}$ and using Lemma 4.7 and Cauchy–Schwarz yields the first line below. For the second line, apply $2ab \le \varepsilon a^2 + \varepsilon^{-1} b^2$, valid for real a, b and $\varepsilon > 0$, with the scalar choices $a := \sqrt{\mathsf{R}_\beta}$, $b := \Delta\sqrt{\mathsf{E}_{\beta\gamma}}$, and $\varepsilon := \kappa t^2 d_e$:

$$\begin{aligned} \mathbb{E}_\mu[w_\beta D g^2] &\le 4\mathsf{R}_\beta + 2\Delta\sqrt{\mathsf{R}_\beta \mathsf{E}_{\beta\gamma}} + \Delta\mathsf{E}_{\beta\gamma} \\ &\le (4 + \kappa t^2 d_e)\mathsf{R}_\beta + \left(\frac{\Delta^2}{\kappa t^2 d_e} + \Delta\right)\mathsf{E}_{\beta\gamma} \\ &\le (4 + \kappa t^2 d_e)\mathsf{R}_\beta + \frac{1001q^2}{t^2 d_e}\mathsf{E}_{\beta\gamma}. \end{aligned}$$

The last step uses $\kappa = 10^{-3}$, $\Delta \le q$, and $t^2 d_e \le q$. □

Proof of Lemma 4.3. Fix g and θ as in the statement, and use the quantities $\mathsf{R}, \mathsf{R}_\beta, \mathsf{E}_{\beta\gamma}$ defined above. Define the positive integer $n := q-2$, the number of colors outside p. For every $\beta \notin p$ and $\sigma \in \Omega$,

$$\sum_{\gamma \notin p \cup \{\beta\}} \big((K_\beta(S_\beta))(\sigma) - (K_\gamma(S_\beta))(\sigma)\big) \geq n(K_\beta(S_\beta))(\sigma) - \Delta|S_\beta(\sigma)|.$$

In the following sums, (β, γ) ranges over ordered pairs of distinct colors outside p. The scalar bounds

$$\sum_{\beta,\gamma} \mathsf{R}_\beta \leq n\Delta\mathsf{R}, \qquad \sum_{\beta,\gamma} \mathsf{E}_{\beta\gamma} \leq 2\mathcal{E}_{\mathrm{aux},p}(g)$$

follow from $|B_u(\sigma)| \leq \Delta$, $0 \leq w_\beta(\sigma) \leq 1$, and $\mathrm{Id} - \Pi_r \preceq Q_r$, where $r = \{\beta, \gamma\}$. Each unordered color pair occurs at most twice. Define the cardinality functions $\mathsf{B}_u, \mathsf{B}_v : \Omega \to \mathbb{Z}_{\geq 0}$ by $\mathsf{B}_u(\sigma) := |B_u(\sigma)|$ and $\mathsf{B}_v(\sigma) := |B_v(\sigma)|$. Summing the bounds on $D_{\beta\gamma}$, dividing by n, and substituting into the bound of Lemma 4.4 gives

$$\begin{aligned} \mathbb{E}_\mu[\theta\, U g^2] \geq \Big(1 - \frac{\Delta}{q-2}\Big)\, \mathbb{E}_\mu[\theta \mathsf{B}_u \mathsf{B}_v g^2] \\ - (5\Delta + \kappa t^2 d_e \Delta)\mathsf{R} - \frac{\mathsf{C}q}{t^2 d_e}\mathcal{E}_{\mathrm{aux},p}(g). \end{aligned} \tag{4.8}$$

Define the subset $W := \{\sigma \in \Omega \mid H(\sigma) \leq 2h_e\}$ of Ω. For every $\sigma \in D_{p,e} \cap W$,

$$\begin{aligned} &|B_u(\sigma)| \geq tq - H(\sigma) - 1 \geq tq/2, \\ &|B_v(\sigma)| \geq d_e - H(\sigma) - 1 \geq d_e/2, \qquad 1 - \frac{\Delta}{q-2} \geq \frac{t}{2}, \end{aligned}$$

provided qt^3 is sufficiently large. These follow from $H(\sigma) \leq 2\kappa t^2 d_e$, $d_e \geq tq$, and $d_e \leq q$. Since θ is supported on W, the first term on the right of (4.8) is at least $t^2 q d_e \mathsf{R}/8$.

For V, Lemma 4.4 gives, on the same set,

$$2\,V(\sigma) \leq 2H(\sigma)\Delta \leq 4h_e\Delta \leq 4\kappa t^2 q d_e.$$

Consequently, $\mathbb{E}_\mu[2\theta\, V g^2] \leq 4\kappa t^2 q d_e \mathsf{R}$.

Combining the above estimates, we have

$$\begin{aligned} \mathbb{E}_\mu[\theta(U - 2\,V)g^2] &\geq \left(\frac{1}{8} - 5\kappa - \frac{5}{qt^3}\right) t^2 q d_e \mathsf{R} - \frac{\mathsf{C}q}{t^2 d_e}\mathcal{E}_{\mathrm{aux},p}(g) \\ &\geq \frac{t^2 q d_e}{10}\mathbb{E}_\mu[\theta g^2] - \frac{\mathsf{C}q}{t^2 d_e}\mathcal{E}_{\mathrm{aux},p}(g). \end{aligned}$$

The last inequality holds after increasing K_d, since $\kappa = 10^{-3}$. This proves Lemma 4.3. □

4.2 Strengthening the bound by an exponential weight function

Define the linear operator $L : \mathbb{R}^\Omega \to \mathbb{R}^\Omega$ by

$$L := \sum_{r \cap p = \varnothing} (P_r - \mathrm{Id}).$$

Its entries are

$$L(\sigma, \omega) = \sum_{r \cap p = \varnothing} K_r(\sigma, \omega) \quad (\omega \neq \sigma), \qquad L(\sigma, \sigma) = -\sum_{\omega \neq \sigma} L(\sigma, \omega).$$

Thus, for every function $F : \Omega \to \mathbb{R}$ and every $\sigma \in \Omega$,

$$(LF)(\sigma) = \sum_{\omega \neq \sigma} L(\sigma, \omega)(F(\omega) - F(\sigma)).$$

The following consequence of reversibility supplies the upper bound that will be combined with Lemma 4.3.

Lemma 4.9. *For every $g : \Omega \to \mathbb{R}$ and every $\phi : \Omega \to \mathbb{R}_{>0}$,*

$$\mathcal{E}_{\mathrm{aux},p}(g) \geq \mathbb{E}_\mu \left[g^2 \frac{-L\phi}{\phi} \right].$$

Proof. Subtracting the expectation from the quadratic form gives

$$\begin{aligned} &\mathcal{E}_{\mathrm{aux},p}(g) - \mathbb{E}_\mu \left[g^2 \frac{-L\phi}{\phi} \right] \\ &\quad = \frac{1}{2} \sum_{\substack{\omega,\tau \in \Omega \\ \omega \neq \tau}} \mu(\omega) L(\omega, \tau) \phi(\omega) \phi(\tau) \left(\frac{g(\omega)}{\phi(\omega)} - \frac{g(\tau)}{\phi(\tau)} \right)^2 \geq 0. \end{aligned}$$

The summands are nonnegative because $L(\omega, \tau) \geq 0$ for $\omega \neq \tau$ and ϕ is positive. □

Proof of Theorem 4.1. Fix $g : \Omega \to \mathbb{R}$ supported on $D_{p,e}$. Define the functions $\phi : \Omega \to (0, \infty)$ and $\zeta : \Omega \to (0, 1)$ by

$$\phi(\sigma) := 2^{-H(\sigma)} + 2^{-3h_e/2}, \qquad \zeta(\sigma) := \frac{2^{-H(\sigma)}}{\phi(\sigma)} = \frac{1}{1 + 2^{H(\sigma) - 3h_e/2}}. \tag{4.9}$$

Updates of color pairs disjoint from p preserve $D_{p,e}$. By Remark 4.2 and the fact that L annihilates constant functions, pointwise on $D_{p,e}$ we have

$$-L\phi = -L(2^{-H}) = \tfrac{1}{4} 2^{-H} (U - 2V).$$

Since g is supported on $D_{p,e}$, Lemma 4.9 gives

$$\mathcal{E}_{\mathrm{aux},p}(g) \geq \mathbb{E}_\mu \left[g^2 \frac{-L\phi}{\phi} \right] = \tfrac{1}{4} \mathbb{E}_\mu [\zeta (U - 2V) g^2]. \tag{4.10}$$

The quotient in this expectation is the function $\sigma \mapsto -(L\phi)(\sigma)/\phi(\sigma)$. Apply Lemma 4.3 with the function $\theta : \Omega \to [0, 1]$ given by $\theta(\sigma) := \zeta(\sigma) \mathbf{1}_{\{H(\sigma) \leq 2h_e\}}$. This is a function of H supported on $\{H \leq 2h_e\}$. For $H(\sigma) \leq h_e$, we have $\zeta(\sigma) \geq 1/2$. For $\sigma \in D_{p,e}$ with $H(\sigma) > 2h_e$, the bound $V(\sigma) \leq qH(\sigma)$ and the decrease of the real function $s \mapsto s 2^{-s}$ on $[2, \infty)$ give

$$\zeta(\sigma) V(\sigma) \leq qH(\sigma) 2^{3h_e/2 - H(\sigma)} \leq 2q h_e \, 2^{-h_e/2},$$

where $2h_e \geq 2$ for sufficiently large qt^3. Since $U \geq 0$, splitting the expectation in (4.10) according to $\{H \leq 2h_e\}$ gives

$$\begin{aligned} \mathcal{E}_{\mathrm{aux},p}(g) \geq \frac{t^2 q d_e}{80} &\mathbb{E}_\mu [g^2 \mathbf{1}_{\{H \leq h_e\}}] \\ &- \frac{\mathsf{C} q}{t^2 d_e} \mathcal{E}_{\mathrm{aux},p}(g) - q h_e 2^{-h_e/2} \|g\|_\mu^2. \end{aligned}$$

Rearrange this inequality and use $t^2 d_e \le q$ and $h_e = \kappa t^2 d_e$ to obtain

$$\mathbb{E}_\mu[g^2 \mathbf{1}_{\{H \le h_e\}}] \le \frac{\mathsf{C}}{t^4 d_e^2} \mathcal{E}_{\mathrm{aux},p}(g) + \mathsf{C} e^{-\mathsf{c} t^2 d_e} \|g\|_\mu^2.$$

Since $d_e \ge tq$, this gives

$$\mathbb{E}_\mu[g^2 \mathbf{1}_{\{H \le h_e\}}] \le \frac{\mathsf{C}}{q^2 t^6} \mathcal{E}_{\mathrm{aux},p}(g) + \mathsf{C} e^{-\mathsf{c} q t^3} \|g\|_\mu^2.$$

Choose K_a large enough for the preceding estimates to hold and A to dominate the universal constants on the right. Since $h \le h_e$, the same bound holds with h in place of h_e. □

5 Bounding $\mathcal{E}_S(f)$ by $\mathcal{E}_T(f)$

We prove Theorem 3.1.

Fix a nonzero function $f : \Omega \to \mathbb{R}$ and a scalar $\lambda \in (0, 1]$ satisfying $Sf = \lambda f$. Define the scalars

$$\mathsf{S} \coloneqq \mathcal{E}_S(f), \qquad \mathsf{T} \coloneqq \mathcal{E}_T(f)$$

for this fixed eigenfunction f. Define the integer cutoff

$$\ell_* \coloneqq \lceil 32/(\kappa t^3) \rceil. \tag{5.1}$$

Notice that $\ell_* = O(t^{-3})$. A path $C \subseteq E$ is called *short* if $|C| \le \ell_*$ and *long* if $|C| > \ell_*$. We bound the contributions to S from short paths in Section 5.2, from long paths and cycles in Section 5.3, and from colorings with too few missing colors in Section 5.4. These estimates are combined to prove Theorem 3.1 at the end of this section. Recall from (4.3) that

$$\kappa \coloneqq 10^{-3}, \qquad h_e \coloneqq \kappa t^2 d_e, \qquad d_e \coloneqq \max\{q - \deg(u), q - \deg(v)\} \quad (e = \{u, v\}), \qquad h \coloneqq \kappa t^3 q.$$

The motivation for the cutoff is to ensure $2\Delta/(\ell_* h) \le 1/16$, which is used in (5.12) to bound the total contribution of cuts producing a fixed component.

5.1 Decomposition over components

Lemma 5.1. *There exist self-adjoint linear maps $F_{p,i}$ on $\mathbb{R}^\Omega$, indexed by $p \in \mathcal{P}$ and $i \in E$, such that*

$$\sum_{p,i} \|F_{p,i} g\|_\mu^2 = \mathcal{E}_S(g) \quad \text{for every } g : \Omega \to \mathbb{R}, \qquad \sum_{p,i} \mathcal{E}_{\mathrm{aux},p}(F_{p,i} f) \le 2\mathsf{S}.$$

The construction of these operators and the proof of Lemma 5.1 are given in Appendix A. We use the notation from that appendix: $C_{p,i}(\sigma)$ is the p-Kempe component with minimum edge i in a fixed total order on E, or the empty set if no such component exists. The family $B_p \coloneqq \{b_p(\sigma) \mid \sigma \in \Omega\}$ is the partition into coloring classes. A superscript $\Xi \in B_p$ denotes restriction to that class. We write C_i for the common value of $C_{p,i}$ on Ξ, and similarly use $M_{p,e}(\Xi)$ and $H_{p,e}(\Xi)$ for the common values of these maps there. Restricted norms follow the normalization convention in Section 2.1.

Definition 5.2 (Component signs and product functions)**.** Fix $p \in \mathcal{P}$ and a coloring class $\Xi \in B_p$. Write $I \coloneqq \{i \in E \mid C_i \ne \varnothing\}$ for its component labels and $k \coloneqq |I|$. Choose a reference coloring $\sigma_\Xi^0 : E \to [q]$ with $\sigma_\Xi^0 \in \Xi$. For $i \in I$, define the sign map $\xi_i^\Xi : \Xi \to \{-1, 1\}$ by

$$\xi_i^\Xi(\sigma) \coloneqq \begin{cases} 1, & \sigma|_{C_i} = \sigma_\Xi^0|_{C_i}, \\ -1, & \sigma|_{C_i} \ne \sigma_\Xi^0|_{C_i}, \end{cases} \qquad \sigma \in \Xi,$$

where C_i is the component with label i on Ξ. For each subset $J \subseteq I$, define $\psi_J^\Xi : \Xi \to \mathbb{R}$ by

$$\psi_J^\Xi(\sigma) := \prod_{i \in J} \xi_i^\Xi(\sigma), \qquad \psi_\varnothing^\Xi(\sigma) = 1.$$

We call ξ_i^Ξ the *component signs* and ψ_J^Ξ the *product functions* on Ξ.

Definition 5.3 (Local component operators)**.** Fix $p \in \mathcal{P}$ and a coloring class $\Xi \in B_p$. For each $i \in E$, let C_i be the common value of $C_{p,i}$ on Ξ, and define the linear operator $R_{p,i}^\Xi$ on $\mathbb{R}^\Xi$ by

$$(R_{p,i}^\Xi g)(\sigma) := g(\Theta_{p,C_i}\sigma), \qquad g : \Xi \to \mathbb{R}, \quad \sigma \in \Xi.$$

With Id_Ξ denoting the identity on $\mathbb{R}^\Xi$, set

$$A_{p,i}^\Xi := \frac{\mathrm{Id}_\Xi - R_{p,i}^\Xi}{2}, \qquad N_p^\Xi := \sum_{i \in E} A_{p,i}^\Xi, \qquad F_{p,i}^\Xi := A_{p,i}^\Xi (N_p^\Xi)^{-1/2}. \tag{5.2}$$

The operators $R_{p,i}^\Xi$ are orthogonal, commute, and satisfy $(R_{p,i}^\Xi)^2 = \mathrm{Id}_\Xi$. Hence each $A_{p,i}^\Xi$ is an orthogonal projection and N_p^Ξ is positive semidefinite with respect to the restriction of μ to Ξ. Here $(N_p^\Xi)^{-1/2}$ acts by $a^{-1/2}$ on an eigenspace with eigenvalue $a > 0$ and by zero on the kernel. Also define the averaging operator P_p^Ξ and its complement Q_p^Ξ on $\mathbb{R}^\Xi$ by

$$(P_p^\Xi g)(\sigma) := \frac{1}{|\Xi|} \sum_{\tau \in \Xi} g(\tau), \qquad Q_p^\Xi := \mathrm{Id}_\Xi - P_p^\Xi.$$

Lemma 5.4. *For every $p \in \mathcal{P}$ and $\Xi \in B_p$, consider $I, k, \sigma_\Xi^0, \xi_i^\Xi, \psi_J^\Xi$ from Definition 5.2. Then $|\Xi| = 2^k$, and the functions $\{\psi_J^\Xi \mid J \subseteq I\}$ form an orthogonal basis of $\mathbb{R}^\Xi$, with*

$$\frac{1}{\mathsf{Z}} \sum_{\sigma \in \Xi} \psi_J^\Xi(\sigma)\psi_K^\Xi(\sigma) = \frac{|\Xi|}{\mathsf{Z}} \mathbf{1}_{\{J=K\}}, \qquad J, K \subseteq I.$$

The operators in Definition 5.3 act on this basis as follows, for every $i \in E$ and $J \subseteq I$:

$$R_{p,i}^\Xi \psi_J^\Xi = (-1)^{\mathbf{1}_{\{i \in J\}}} \psi_J^\Xi, \quad A_{p,i}^\Xi \psi_J^\Xi = \mathbf{1}_{\{i \in J\}} \psi_J^\Xi,$$
$$N_p^\Xi \psi_J^\Xi = |J| \psi_J^\Xi, \qquad F_{p,i}^\Xi \psi_J^\Xi = \begin{cases} \dfrac{\mathbf{1}_{\{i \in J\}}}{\sqrt{|J|}} \psi_J^\Xi, & J \neq \varnothing, \\ 0, & J = \varnothing. \end{cases}$$

In particular, $R_{p,i}^\Xi 1 = 1$, whereas $A_{p,i}^\Xi 1 = N_p^\Xi 1 = F_{p,i}^\Xi 1 = 0$. The resulting identities and inequalities of local operators and norms are

$$\sum_{i \in E} (F_{p,i}^\Xi)^2 = Q_p^\Xi, \qquad \|F_{p,i}^\Xi g\|_{\mu,\Xi} \le \|A_{p,i}^\Xi g\|_{\mu,\Xi} \le \|g\|_{\mu,\Xi} \quad \text{for every } g : \Xi \to \mathbb{R}.$$

The proof is given in Appendix A.

5.2 Flipping short paths

Lemma 5.5. *For each $p \in \mathcal{P}$, define the set-valued map $I_p : \Omega \to 2^E$ by*

$$I_p(\sigma) \coloneqq \{i \in E \mid C_{p,i}(\sigma) \text{ is a nonempty short path and } H_{p,e}(\sigma) > h_e \text{ for every } e \in C_{p,i}(\sigma)\}.$$

Define the scalar

$$\mathsf{S}_{\mathrm{s}} \coloneqq \sum_{p,i} \mathbb{E}_\mu \left[\mathbf{1}_{\{i \in I_p\}} (F_{p,i} f)^2 \right],$$

where the indicator is the map $\sigma \mapsto \mathbf{1}_{\{i \in I_p(\sigma)\}}$. There exist positive universal constants $\mathsf{C}, \mathsf{K}_{\mathrm{s}}$ such that, for every $qt^3 > \mathsf{K}_{\mathrm{s}}$, we have

$$\mathsf{S}_{\mathrm{s}} \le \mathsf{C} t^{-8} \mathsf{T}. \tag{5.3}$$

Here $h_e \coloneqq \kappa t^2 \max\{q - \deg(u), q - \deg(v)\}$ for $e = \{u, v\}$, as in (4.3).

Lemma 5.6. *There exists a positive universal constant $\mathsf{K}_{\mathrm{short}}$ such that the following holds for every $qt^3 > \mathsf{K}_{\mathrm{short}}$. For every $\sigma \in \Omega$ and $i \in I_p(\sigma)$, put $\ell \coloneqq |C_{p,i}(\sigma)|$. There exists a random sequence $\Gamma_{\sigma,p,i}$ of $2\ell + 1$ colorings starting at σ and ending at $\Theta_{p,C_{p,i}(\sigma)}\sigma$, with every consecutive pair a legal single-edge recoloring.*

Proof. Fix $\sigma \in \Omega$ and $i \in I_p(\sigma)$, and define $C_i \coloneqq C_{p,i}(\sigma) \subseteq E$ and the scalar $\ell \coloneqq |C_i|$. Choose one of the two directions along this path uniformly at random and label its edges $e_1, \dots, e_\ell$ in that direction. Choose β_1 uniformly from $M_{p,e_1}(\sigma)$, then, for $j = 2, \dots, \ell$, choose β_j uniformly from $M_{p,e_j}(\sigma) \setminus \{\beta_{j-1}\}$ conditional on the previous choices. These sets are nonempty because $H_{p,e}(\sigma) > h_e$ on C_i and qt^3 is sufficiently large. We call $\beta_1, \dots, \beta_\ell$ the *auxiliary colors*.

Using the notation $\omega^{e \leftarrow c}$ from Section 2, define a recoloring sequence as follows. Define the sequence $(\omega_0, \dots, \omega_{2\ell})$ recursively by

$$\begin{aligned} \omega_0 &\coloneqq \sigma, & \omega_1 &\coloneqq \omega_0^{e_1 \leftarrow \beta_1}, \\ \omega_{2j-2} &\coloneqq \omega_{2j-3}^{e_j \leftarrow \beta_j}, & \omega_{2j-1} &\coloneqq \omega_{2j-2}^{e_{j-1} \leftarrow s_p(\sigma(e_{j-1}))} \quad (2 \le j \le \ell), \\ \omega_{2\ell} &\coloneqq \omega_{2\ell-1}^{e_\ell \leftarrow s_p(\sigma(e_\ell))}. \end{aligned}$$

At the first step, we assign β_1 to e_1. At each even step before the last, we assign the next edge its auxiliary color. At each subsequent odd step, we replace the preceding edge's auxiliary color by its original color exchanged by s_p. The final step makes this exchange on the last edge. Each auxiliary color is missing at both endpoints of its edge in σ, adjacent auxiliary colors differ, and the edges carrying them separate the exchanged colors from the original colors. Thus every ω_j belongs to Ω, each step is a legal single-edge recoloring, and $\omega_{2\ell} = \Theta_{p,C_i}\sigma$. Defining the random sequence $\Gamma_{\sigma,p,i} \coloneqq (\omega_0, \dots, \omega_{2\ell})$ proves Lemma 5.6. □

Recall the arc set A from Section 2. For $d = (\omega, \tau, e) \in A$, define $N_d : \bigcup_{n=1}^{2\ell_*} \Omega^{n+1} \to \mathbb{N}_0$ by

$$N_d(\eta_0, \dots, \eta_n) \coloneqq \sum_{j=1}^{n} \mathbf{1}_{\{\eta_{j-1} = \omega,\ \eta_j = \tau\}}.$$

Thus $N_d(\Gamma)$ is the scalar number of occurrences of d in Γ. Define the nonnegative scalars

$$\mathsf{R}_d \coloneqq \sum_{\sigma,p} \sum_{i \in I_p(\sigma)} \mathbb{E}[N_d(\Gamma_{\sigma,p,i})], \qquad \mathsf{R} \coloneqq \max_{d \in A} \mathsf{R}_d, \tag{5.4}$$

where the expectation is over the random choices in Lemma 5.6. We call R the *congestion* of these sequences.

Lemma 5.7. *There exist positive universal constants* $\mathsf{C}, \mathsf{K}_{\mathrm{cong}}$ *such that, for every* $qt^3 > \mathsf{K}_{\mathrm{cong}}$*, the random sequences constructed in Lemma 5.6 satisfy* $\mathsf{R} \le \mathsf{C}t^{-5}$.

Proof. Fix $d = (\omega, \tau, e) \in A$, and write $e = \{u, v\}$, $a \coloneqq \omega(e)$, $b \coloneqq \tau(e)$, and $d_z \coloneqq q - \deg(z)$ for $z \in \{u, v\}$. Let J_d be the set of tuples $(\sigma, p, i, \vec{C})$, where $\sigma \in \Omega$, $p \in \mathcal{P}$, $i \in I_p(\sigma)$, and $\vec{C}$ is a traversal direction of $C_{p,i}(\sigma)$, for which the random sequence contains d with positive probability conditional on that direction. We define an injective map

$$\Phi_d : J_d \longrightarrow \{a, b\} \times ([q] \setminus \{a, b\}) \times \Omega \times \{u, v\} \times E.$$

For $(\sigma, p, i, \vec{C}) \in J_d$, let γ be the unique color in $p \cap \{a, b\}$ and write $p = \{\gamma, c\}$. Label the path in direction $\vec{C}$ as $x_0, e_1, x_1, \ldots, e_\ell, x_\ell$, with $e_j = \{x_{j-1}, x_j\}$, and let $e = e_j$. Set $\Phi_d(\sigma, p, i, \vec{C}) \coloneqq (\gamma, c, \eta, z, g)$, where

γ	type of recoloring	η	z	g
a	assignment of β_j	ω	x_{j-1}	e_{j-1}
b	assignment of $s_p(\sigma(e_j))$	τ	x_j	e_{j+1}

We use $g = e$ when the indicated neighboring edge is absent, namely when $j = 1$ and $a \in p$, or $j = \ell$ and $b \in p$. Otherwise, g is incident to z and is the unique edge of $C_{p,i}(\sigma) \setminus \{e\}$ whose color in η lies outside p. These choices depend only on $(\sigma, p, i, \vec{C})$.

To prove injectivity, recover $p = \{\gamma, c\}$ and let C be the component containing e in the graph with edge set $E_p(\eta) \cup \{g\}$. Then $C = C_{p,i}(\sigma)$ and $i = \min C$. The direction enters e at z if $a \in p$ and leaves e at z if $b \in p$, which determines $\vec{C}$ and j. Outside C, σ agrees with η. On C, it is the unique alternating p-coloring with $\sigma(e) = a$ if $a \in p$ and $\sigma(e) = c$ if $b \in p$. Thus the tuple determines $(\sigma, p, i, \vec{C})$, proving that Φ_d is injective.

For every tuple in the image, properness gives

$$c \notin \{a, b\} \cup \{\omega(f) \mid f \neq e,\ z \in f\},$$

so there are at most d_z choices for c. For fixed γ and z, the coordinate η is fixed, and g has one choice $g = e$ and at most Δ other choices. For $f = e$ and also for $f = g$ when $g \neq e$, we have $z \in f$ and

$$|M_{p,f}(\sigma)| > h_f = \kappa t^2 d_f \ge \kappa t^2 d_z.$$

After increasing $\mathsf{K}_{\mathrm{cong}}$, each prescribed auxiliary color has conditional probability at most $\alpha_z \coloneqq 2/(\kappa t^2 d_z)$, since its sampling set excludes at most one color. The occurrence of d prescribes β_j, and, when $g \neq e$, also β_{j-1} if $a \in p$ or β_{j+1} if $b \in p$. Conditional on $\vec{C}$, its probability is therefore at most α_z when $g = e$ and α_z^2 otherwise.

For each tuple in J_d, the arc d can occur only once: its type and the position of e in the traversal fix its step. Each direction has probability $1/2$. Using the injection and summing over the two choices of γ, we obtain

$$\begin{aligned}
\mathsf{R}_d &= \frac{1}{2} \sum_{(\sigma,p,i,\vec{C}) \in J_d} \Pr[N_d(\Gamma_{\sigma,p,i}) = 1 \mid \vec{C}] \\
&\le \sum_{z \in \{u,v\}} d_z \left(\alpha_z + \Delta \alpha_z^2 \right) \\
&= \sum_{z \in \{u,v\}} \left(\frac{2}{\kappa t^2} + \frac{4\Delta}{\kappa^2 t^4 d_z} \right) \le \frac{4}{\kappa t^2} + \frac{8}{\kappa^2 t^5} \le \mathsf{C} t^{-5},
\end{aligned} \tag{5.5}$$

where $d_z \ge tq$ and $\Delta \le q$. Taking the maximum over $d \in A$ proves Lemma 5.7. □

Proof of Lemma 5.5. The set $\{\sigma \mid i \in I_p(\sigma)\}$ is a union of classes $\Xi \in B_p$. On each such class, Lemma 5.4 gives

$$\begin{aligned}\|F^{\Xi}_{p,i}(f|_\Xi)\|^2_{\mu,\Xi} &\le \|A^{\Xi}_{p,i}(f|_\Xi)\|^2_{\mu,\Xi} \\ &= \frac{1}{4}\sum_{\sigma\in\Xi}\mu(\sigma)\big(f(\sigma)-f(\Theta_{p,C_i}\sigma)\big)^2.\end{aligned}$$

For $\sigma \in \Omega$, $p \in \mathcal{P}$, and $i \in I_p(\sigma)$, write $C_i := C_{p,i}(\sigma)$, $\ell := |C_i|$, and $\Gamma_{\sigma,p,i} = (\omega_0, \dots, \omega_{2\ell})$. Telescoping and Cauchy–Schwarz give

$$\begin{aligned}\big(f(\sigma)-f(\Theta_{p,C_i}\sigma)\big)^2 &= \left(\sum_{j=1}^{2\ell}\big(f(\omega_{j-1})-f(\omega_j)\big)\right)^2 \\ &\le 2\ell\sum_{j=1}^{2\ell}\big(f(\omega_{j-1})-f(\omega_j)\big)^2.\end{aligned}$$

Taking expectations, using $\ell \le \ell_*$ and the uniformity of μ, and summing yields

$$\begin{aligned}\mathsf{S}_{\mathrm{s}} &\le \frac{1}{4}\sum_{\sigma,p}\sum_{i\in I_p(\sigma)}\mu(\sigma)\big(f(\sigma)-f(\Theta_{p,C_i}\sigma)\big)^2 \\ &\le \frac{\ell_*}{2}\sum_{\sigma,p}\sum_{i\in I_p(\sigma)}\mu(\sigma)\mathbb{E}\left[\sum_{d=(\omega,\tau,e)\in A}N_d(\Gamma_{\sigma,p,i})\big(f(\omega)-f(\tau)\big)^2\right] \\ &= \frac{\ell_*}{2}\sum_{d=(\omega,\tau,e)\in A}\mu(\omega)\mathsf{R}_d\big(f(\omega)-f(\tau)\big)^2 \\ &\le 2\ell_*\mathsf{R}\mathsf{T} \le \mathsf{C}t^{-8}\mathsf{T},\end{aligned}$$

by (2.2), Lemma 5.7, and $\ell_* = O(t^{-3})$. □

5.3 Removing an edge from a long path or a cycle

Lemma 5.8. *Define the scalar*

$$\mathsf{S}_{\mathrm{c}} := \sum_{p\in\mathcal{P}}\sum_{\Xi\in B_p}\sum_{\substack{i\in I_{p,\Xi}\\ C_i \text{ is a long path or a cycle}}}\frac{|\{e\in C_i \mid H_{p,e}(\Xi) > h\}|}{|C_i|}\|F^{\Xi}_{p,i}(f|_\Xi)\|^2_{\mu,\Xi}.$$

There exist positive universal constants $\mathsf{C}, \mathsf{K}_{\mathrm{c}}$ *such that, for every* $qt^3 > \mathsf{K}_{\mathrm{c}}$*, we have*

$$\mathsf{S}_{\mathrm{c}} \le \frac{1}{4}\mathsf{S} + \mathsf{C}t^{-3}\mathsf{T}. \tag{5.6}$$

For a long path or a cycle C_i, we compare $\|F_{p,i}f\|^2_\mu$ with norms associated with the components of $C_i \setminus \{e\}$, using a recoloring of e to a color outside p. Use the family B_p of coloring classes introduced in Section 5.1. The sets C_i, $M_{p,e}$ and the values $H_{p,e}$ are constant on $\Xi \in B_p$. A notation such as $H_{p,e}(\Xi)$ denotes this common scalar value. Define the finite index set A_{cut} to consist of all tuples $\alpha = (p, \Xi, i, e, \beta)$ satisfying

$$\Xi \in B_p, \quad C_i \text{ is a long path or a cycle}, \quad e \in C_i, \quad H_{p,e}(\Xi) > h, \quad \beta \in M_{p,e}(\Xi).$$

For every $\alpha \in A_{\mathrm{cut}}$, define the real scalar

$$w_\alpha \coloneqq \frac{1}{|C_i| H_{p,e}(\Xi)}, \qquad 0 < w_\alpha \le \frac{1}{4h}, \tag{5.7}$$

and call w_α the *cut weight*. It is the probability of this particular choice of e, β when e is uniform on C_i and, for $H_{p,e} > h$, β is uniform on $M_{p,e}$. The upper bound uses $|C_i| \ge 4$, because C_i is a long path or an even cycle in a simple graph.

Define the recoloring map $\Phi_\alpha : \Xi \to \Omega$ by

$$\Phi_\alpha(\omega) \coloneqq \omega^{e \leftarrow \beta}.$$

Its image lies in a single class $\Xi_1 \coloneqq b_p(\Phi_\alpha(\omega)) \in B_p$, independent of $\omega \in \Xi$. We henceforth regard $\Phi_\alpha : \Xi \to \Xi_1$. This map is injective: the edges of $C_i \setminus \{e\}$ adjacent to e all have the same color in p, and the other color of p uniquely restores e. Norms on Ξ_1 use the entire class, including states outside $\Phi_\alpha(\Xi)$, with mass $1/\mathsf{Z}$ per coloring.

For any $\tau \in \Xi_1$, define the set

$$W \coloneqq \{D \in K_p(\tau) \mid D \subseteq C_i \setminus \{e\}\} \subseteq 2^E.$$

This set is independent of τ. Its elements are precisely the nonempty edge sets of the components of $(V, C_i \setminus \{e\})$, and $1 \le |W| \le 2$. For such a component D, define the local flip operator $R'_D : \mathbb{R}^{\Xi_1} \to \mathbb{R}^{\Xi_1}$ by $R'_D \coloneqq R^{\Xi_1}_{p,\min D}$. Define the linear operators $R_W, A_W, T_\alpha : \mathbb{R}^{\Xi_1} \to \mathbb{R}^{\Xi_1}$ by

$$R_W \coloneqq \prod_{D \in W} R'_D, \qquad A_W \coloneqq \frac{\mathrm{Id}_{\Xi_1} - R_W}{2}, \qquad T_\alpha \coloneqq A_W (N_p^{\Xi_1})^{-1/2}.$$

The factors R'_D commute, so their product is independent of order. Likewise define $F'_{p,D} : \mathbb{R}^{\Xi_1} \to \mathbb{R}^{\Xi_1}$ by $F'_{p,D} \coloneqq F^{\Xi_1}_{p,\min D}$.

Lemma 5.9. *For every $\alpha \in A_{\mathrm{cut}}$,*

$$\begin{aligned} F^{\Xi}_{p,i}(f \circ \Phi_\alpha) &= (T_\alpha(f|_{\Xi_1})) \circ \Phi_\alpha, \\ T_\alpha^2 &\preceq \sum_{D \in W} (F'_{p,D})^2. \end{aligned} \tag{5.8}$$

The first identity is an equality of maps $\Xi \to \mathbb{R}$. The inequality holds in the quadratic-form sense on $\mathbb{R}^{\Xi_1}$. Moreover,

$$\|F^{\Xi}_{p,i}(f|_\Xi)\|_{\mu,\Xi} \le \|T_\alpha(f|_{\Xi_1})\|_{\mu,\Xi_1} + \|f|_\Xi - f \circ \Phi_\alpha\|_{\mu,\Xi}. \tag{5.9}$$

Proof. For $\tau \in \Xi_1$ and $J \subseteq K_p(\tau)$, consider the product basis function $\psi : \Xi_1 \to \mathbb{R}$ defined by $\psi \coloneqq \prod_{D \in J} \xi_D$, where the sign map $\xi_D : \Xi_1 \to \{-1, 1\}$ is $\xi_D \coloneqq \xi^{\Xi_1}_{\min D}$ from Definition 5.2. Define the scalars

$$k \coloneqq |\{D \in J \mid D \notin W\}|, \qquad j \coloneqq |\{D \in J \mid D \in W\}|.$$

Both $F^{\Xi}_{p,i}(\psi \circ \Phi_\alpha)$ and $(T_\alpha \psi) \circ \Phi_\alpha$ vanish for even j. For odd j, necessarily $j = 1$, and both functions equal $(\psi \circ \Phi_\alpha)/\sqrt{k+1}$. These identities on the product basis give the first identity of Lemma 5.9. If $k + j > 0$, the multipliers of T_α^2 and $\sum_{D \in W} (F'_{p,D})^2$ on ψ are, respectively,

$$\frac{\mathbf{1}_{\{j \text{ is odd}\}}}{k+j} \le \frac{j}{k+j}.$$

Both multipliers are zero when $k + j = 0$. This gives the operator inequality in Lemma 5.9. The contraction property of $F^{\Xi}_{p,i}$ and injectivity of Φ_α then give the norm bound in Lemma 5.9 by the triangle inequality. □

Lemma 5.10. *There exist positive universal constants* $\mathsf{C}, \mathsf{K}_{\mathrm{sum}}$ *such that, for every* $qt^3 > \mathsf{K}_{\mathrm{sum}}$*, the cut weights and operators satisfy*

$$\sum_{\alpha} w_\alpha \|T_\alpha(f|_{\Xi_1})\|^2_{\mu,\Xi_1} \le \mathsf{S}/8, \tag{5.10}$$

$$\sum_{\alpha} w_\alpha \|f|_{\Xi} - f \circ \Phi_\alpha\|^2_{\mu,\Xi} \le \frac{4 \cdot 2\Delta}{4h}\mathsf{T} \le \mathsf{C}t^{-3}\mathsf{T}. \tag{5.11}$$

Proof. To sum $w_\alpha \|T_\alpha(f|_{\Xi_1})\|^2_{\mu,\Xi_1}$, fix p, Ξ_1, and a component $D \in W$. If C_i is a long path, then e is incident to an endpoint of D. There are at most 2Δ possibilities for e, each with $w_\alpha \le 1/(\ell_* h)$. If C_i is a cycle, e joins the two endpoints of D. Simplicity of G allows at most one such edge, with $w_\alpha \le 1/(4h)$. Therefore the sum of the cut weights producing this fixed component D is at most

$$\frac{2\Delta}{\ell_* h} + \frac{1}{4h} \le \frac{1}{8} \tag{5.12}$$

for sufficiently large qt^3, by (5.1). For each e, its color in Ξ_1 determines β, and adding e to the two-color subgraph determines C_i and Ξ. Moreover,

$$M_{p,e}(\Xi) = \{\beta\} \cup M_{p,e}(\Xi_1).$$

Thus w_α is independent of the component orientations in Ξ_1. Lemmas 5.1 and 5.9 give the first bound in Lemma 5.10, where all sums over α are over A_{cut}.

For the difference term in the norm bound of Lemma 5.9, fix a recoloring $(\omega, \tau, e) \in A$. One color of p is $\omega(e)$. The other must occur on an edge adjacent to e because $|C_i| > 1$. There are at most 2Δ possibilities for this other color. Combining this bound with $w_\alpha \le 1/(4h)$ and the normalization in (2.2) yields the second bound in Lemma 5.10. □

Proof of Lemma 5.8. Applying $(a+b)^2 \le 2a^2 + 2b^2$ to the norm bound in Lemma 5.9, multiplying by w_α, and summing gives

$$\sum_{\alpha} w_\alpha \|F^{\Xi}_{p,i}(f|_{\Xi})\|^2_{\mu,\Xi} \le \frac{1}{4}\mathsf{S} + \mathsf{C}t^{-3}\mathsf{T}$$

by Lemma 5.10. For fixed p, Ξ, i, e satisfying $H_{p,e} > h$,

$$\sum_{\beta \in M_{p,e}(\Xi)} w_{(p,\Xi,i,e,\beta)} = \frac{1}{|C_i|}.$$

Consequently, the sum on the left is exactly S_{c}. □

5.4 Colorings with too few missing colors

Lemma 5.11. *For every* $qt^3 > \mathsf{K}_a$*, the real scalar* $\mathsf{S}_{\mathrm{b}} \coloneqq \mathsf{S} - \mathsf{S}_{\mathrm{s}} - \mathsf{S}_{\mathrm{c}}$ *satisfies*

$$0 \le \mathsf{S}_{\mathrm{b}} \le (\ell_* + 1)\left(\rho\mathsf{S} + \frac{\mathsf{A}}{q^2t^6}\sum_{p,i}\mathcal{E}_{\mathrm{aux},p}(F_{p,i}f)\right), \qquad \rho \coloneqq \mathsf{A}e^{-\mathsf{c}qt^3}, \tag{5.13}$$

where A, c *are the constants in Theorem 4.1.*

Lemma 5.12. *Assume $qt^3 > \mathsf{K}_a$, and fix $p \in \mathcal{P}$ and $i \in E$. Let $w_e : \Omega \to [0,1]$, $e \in E$, be weights unchanged under every update of a color pair disjoint from p, each supported on $\{\sigma \in \Omega \mid e \in C_{p,i}(\sigma)\}$, and assume $\sum_e w_e(\sigma) \le b$ for every $\sigma \in \Omega$, where $b \ge 0$. Then*

$$\sum_e \mathbb{E}_\mu[w_e (F_{p,i} f)^2 \mathbf{1}_{\{H_{p,e} \le h_e\}}] \le b \left(\rho \|F_{p,i} f\|_\mu^2 + \frac{\mathsf{A}}{q^2 t^6} \mathcal{E}_{\mathrm{aux},p}(F_{p,i} f) \right).$$

The same inequality holds with h in place of every h_e.

Proof. For each e, the map $\sqrt{w_e} F_{p,i} f : \Omega \to \mathbb{R}$ is supported on $D_{p,e}$. Apply Theorem 4.1 to this map. Invariance gives $Q_r(\sqrt{w_e} F_{p,i} f) = \sqrt{w_e} Q_r F_{p,i} f$ when $r \cap p = \varnothing$. Sum over e and use $\sum_e w_e \le b$. The final assertion follows from $h \le h_e$. □

Proof of Lemma 5.11. The identity in Lemma 5.1 and the definitions of S_s and S_c give $\mathsf{S}_\mathrm{b} \ge 0$. The estimate in Lemma 5.5 excludes short paths with at least one edge e satisfying $H_{p,e} \le h_e$. The estimate in Lemma 5.8 excludes choices of e in long paths or cycles satisfying $H_{p,e} \le h$. For each p, i, e, define the functions $w^\mathrm{s}_{p,i,e}, w^\mathrm{c}_{p,i,e} : \Omega \to [0,1]$ by

$$\begin{aligned} w^\mathrm{s}_{p,i,e}(\sigma) &:= \mathbf{1}_{\{C_{p,i}(\sigma) \text{ is a short path, } e \in C_{p,i}(\sigma)\}}, \\ w^\mathrm{c}_{p,i,e}(\sigma) &:= \begin{cases} |C_{p,i}(\sigma)|^{-1}, & C_{p,i}(\sigma) \text{ is a long path or a cycle and } e \in C_{p,i}(\sigma), \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \tag{5.14}$$

We call these functions the *short-path edge weight* and the *cut edge weight*, respectively. Both are zero when the component is absent. A union bound over $e \in C_i$ for short paths, and the uniform average over $e \in C_i$ for long paths and cycles, show that the contribution to S not covered by Lemmas 5.5 and 5.8 is at most

$$\mathsf{S}_\mathrm{b} \le \sum_{p,i,e} \mathbb{E}_\mu \Big[(F_{p,i} f)^2 \left(w^\mathrm{s}_{p,i,e} \mathbf{1}_{\{H_{p,e} \le h_e\}} + w^\mathrm{c}_{p,i,e} \mathbf{1}_{\{H_{p,e} \le h\}} \right) \Big]. \tag{5.15}$$

The two weight functions depend only on E_p and therefore are unchanged under updates with $r \cap p = \varnothing$. For $w := w^\mathrm{s}_{p,i,e}$ or $w := w^\mathrm{c}_{p,i,e}$, the function $\sqrt{w}\, F_{p,i} f : \Omega \to \mathbb{R}$ is supported on $\{\sigma \mid e \in E_p(\sigma)\}$. The pointwise bounds

$$\sum_e w^\mathrm{s}_{p,i,e} \le \ell_*, \qquad \sum_e w^\mathrm{c}_{p,i,e} \le 1$$

and Lemma 5.12, summed over p, i, give the bound in Lemma 5.11 by Lemma 5.1. The functions $w^\mathrm{s}_{p,i,e}$ and $w^\mathrm{c}_{p,i,e}$ do not depend on $H_{p,e}$. The indicators involving $H_{p,e}$ remain on the left side of the bound in Theorem 4.1. The cut weights $w_\alpha := 1/(|C_i| H_{p,e})$ from (5.7) are not used in this application of the lemma, because updates with $r \cap p = \varnothing$ may change $H_{p,e}$. □

Proof of Theorem 3.1. By definition, $\mathsf{S} = \mathsf{S}_\mathrm{s} + \mathsf{S}_\mathrm{c} + \mathsf{S}_\mathrm{b}$. Apply Lemmas 5.5, 5.8 and 5.11, together with the energy bound in Lemma 5.1 and $\ell_* = O(t^{-3})$, to obtain

$$\mathsf{S} \le \mathsf{C} t^{-8} \mathsf{T} + \frac{1}{4} \mathsf{S} + \mathsf{C} \left(t^{-3} e^{-\mathsf{c} q t^3} + \frac{1}{q^2 t^9} \right) \mathsf{S}. \tag{5.16}$$

Since $e^{-s} \le 1/s$ for $s > 0$,

$$t^{-3} e^{-\mathsf{c} q t^3} \le \frac{1}{\mathsf{c} q t^6} \le \frac{1}{\mathsf{c} q t^{14}}, \qquad \frac{1}{q^2 t^9} \le \frac{1}{(q t^{14})^2}.$$

Choose K_1 large enough to meet the thresholds in the preceding lemmas and to ensure that, for $qt^{14} \geq \mathsf{K}_1$,

$$\mathsf{C}\left(t^{-3}e^{-\mathsf{c}qt^3} + \frac{1}{q^2t^9}\right) \leq \mathsf{C}\left(\frac{1}{\mathsf{c}qt^{14}} + \frac{1}{(qt^{14})^2}\right) \leq \frac{1}{4}. \tag{5.17}$$

Subtracting $\mathsf{S}/4 + \mathsf{S}/4$ from (5.16) proves Theorem 3.1. □

6 Bounding $\mathcal{E}_T(f)$ by $\mathcal{E}_S(f)$

In this section, we prove Theorem 3.2.

As in Section 5, for $f : \Omega \to \mathbb{R}$ we use the shorthands $\mathsf{S} \coloneqq \mathcal{E}_S(f)$ and $\mathsf{T} \coloneqq \mathcal{E}_T(f)$. Recall the set $A \subseteq \Omega \times \Omega \times E$ of oriented legal recolorings from Section 2. Its elements $d = (\omega, \tau, e)$ are called arcs. Let ν be the uniform measure on A that assigns weight $1/(4\mathsf{Z})$ to each arc. Formally,

$$\nu(U) \coloneqq \frac{|U|}{4\mathsf{Z}} \quad (U \subseteq A).$$

For $\psi : A \to \mathbb{R}$, integration is given by

$$\int_A \psi \, d\nu \coloneqq \frac{1}{4\mathsf{Z}} \sum_{d \in A} \psi(d).$$

Inner products and norms with respect to ν follow the conventions in Section 2.1. Define the linear map ∇ and the function $g : A \to \mathbb{R}$ by

$$\nabla : \mathbb{R}^\Omega \longrightarrow \mathbb{R}^A, \qquad (\nabla f)(\omega, \tau, e) \coloneqq f(\omega) - f(\tau), \qquad g \coloneqq \nabla f.$$

We have $\|g\|_\nu^2 = \mathsf{T}$ by (2.2). Define the maps

$$\begin{aligned} \pi &: A \longrightarrow \mathcal{P}, & \pi(\omega, \tau, e) &\coloneqq \{\omega(e), \tau(e)\}, \\ H &: A \longrightarrow \{0, \dots, q-2\}, & H(\omega, \tau, e) &\coloneqq H_{\pi(\omega,\tau,e),e}(\omega). \end{aligned}$$

Since ω and τ agree outside e, $H_{\pi(\omega,\tau,e),e}(\omega) = H_{\pi(\omega,\tau,e),e}(\tau)$.

Define the nonnegative quadratic form $\mathcal{B}_T : \mathbb{R}^\Omega \to [0, \infty)$ by

$$\mathcal{B}_T(f) \coloneqq \sum_{e,p} \sum_{r \cap p = \varnothing} \|Q_r G_{e,p} f\|_\mu^2. \tag{6.1}$$

The map $G_{e,p}$ was defined in Section 2. On a legal recoloring with color pair p it satisfies $G_{e,p}f(\omega) = (f(\omega) - f(\tau))/2$. Define the scalar

$$\mathsf{Y} \coloneqq \frac{1}{2}\int_A H g^2 \, d\nu + \mathcal{B}_T(f). \tag{6.2}$$

6.1 Coupling the color-pair updates

Lemma 6.1. *For every $r \in \mathcal{P}$, there exist linear maps*

$$M_r : \mathbb{R}^A \to \mathbb{R}^A, \qquad D_r : \mathbb{R}^\Omega \to \mathbb{R}^A,$$

such that M_r is self-adjoint with respect to $\langle \cdot, \cdot \rangle_\nu$ and $M_r^2 = M_r$. With $D \coloneqq \sum_r D_r : \mathbb{R}^\Omega \to \mathbb{R}^A$, every $f : \Omega \to \mathbb{R}$ satisfies

$$\nabla P_r f = M_r \nabla f + D_r f, \qquad \nabla S f = \sum_r (\mathrm{Id} - M_r)\nabla f - Df. \tag{6.3}$$

For each $r \in \mathcal{P}$, we define a linear operator $M_r : \mathbb{R}^A \to \mathbb{R}^A$ to compare $M_r \nabla f$ with $\nabla P_r f$. Use the component collection K_r and flip maps $\Theta_{r,F}$ from Section 2.2. For $F \subseteq E$, define the simultaneous flip map $T_{r,F} : A \to A$ by

$$T_{r,F}(\omega, \tau, e) \coloneqq \begin{cases} (\Theta_{r,F}\omega, \Theta_{r,F}\tau, e), & F \in K_r(\omega) \cap K_r(\tau), \\ (\omega, \tau, e), & \text{otherwise.} \end{cases}$$

Recall that $\pi(d) = \{\omega(e), \tau(e)\}$ is the pair of colors used by $d = (\omega, \tau, e)$ on its recolored edge. When $|r \cap \pi(d)| = 1$, denote the two colorings by ω_a, ω_c, where $\omega_a(e) = a \in r$ and $\omega_c(e) = c \notin r$, and let $C \in K_r(\omega_a)$ be the component containing e. Define $T_{r,0} : A \to A$ by

$$T_{r,0}(\omega, \tau, e) \coloneqq \begin{cases} (\Theta_{r,C}\omega, \Theta_{r,C\setminus\{e\}}\tau, e), & \omega(e) \in r,\ \tau(e) \notin r, \\ (\Theta_{r,C\setminus\{e\}}\omega, \Theta_{r,C}\tau, e), & \omega(e) \notin r,\ \tau(e) \in r, \\ (\omega, \tau, e), & \text{otherwise.} \end{cases}$$

The set $C \setminus \{e\}$ is a union of nonempty r-components of ω_c. There are zero such components if $C = \{e\}$, one if C is a cycle or C is a path with at least two edges and e is an end edge, and two if C is a path and e is an internal edge. Thus $T_{r,0}$ produces legal recolorings in both cases.

For $r \neq \pi(d)$, define the collection $\mathcal{T}_r(d)$ of component-flip maps by

$$\mathcal{T}_r(d) \coloneqq \begin{cases} \{\, T_{r,F} \mid F \in K_r(\omega) \,\}, & r \cap \pi(d) = \varnothing, \\ \{\, T_{r,0} \,\} \cup \{\, T_{r,F} \mid F \in K_r(\omega_a) \setminus \{C\} \,\}, & |r \cap \pi(d)| = 1. \end{cases}$$

Each indexed component F remains an r-Kempe component with the same edge set in both colorings after any of these flips. When $|r \cap \pi(d)| = 1$, the distinguished component C also has the same edge set in the coloring whose color on e lies in r. Consequently, $\mathcal{T}_r(d') = \mathcal{T}_r(d)$ whenever $d' = T(d)$ with $T \in \mathcal{T}_r(d)$. On the arcs obtained from d by compositions, the maps commute, and applying any one twice returns the original arc.

Define $m_r(d)$ by

$$m_r(d) \coloneqq \begin{cases} |K_r(\omega)|, & r \cap \pi(d) = \varnothing, \\ |K_r(\omega_a)|, & |r \cap \pi(d)| = 1. \end{cases}$$

Thus $m_r(d)$ is the number of component-flip maps in $\mathcal{T}_r(d)$. Choose an ordering $T_1, \ldots, T_{m_r(d)}$ of $\mathcal{T}_r(d)$. For a binary vector $\varepsilon \in \{0,1\}^{m_r(d)}$, its jth coordinate ε_j specifies whether T_j is applied: $T_j^0 = \mathrm{Id}$ and $T_j^1 = T_j$. Define

$$\Phi_{r,d}(\varepsilon) \coloneqq T_1^{\varepsilon_1} \cdots T_{m_r(d)}^{\varepsilon_{m_r(d)}} d.$$

For every $\psi : A \to \mathbb{R}$, define M_r by

$$(M_r\psi)(d) \coloneqq \begin{cases} 0, & \pi(d) = r, \\ 2^{-m_r(d)} \displaystyle\sum_{\varepsilon \in \{0,1\}^{m_r(d)}} \psi(\Phi_{r,d}(\varepsilon)), & \pi(d) \neq r. \end{cases}$$

Thus, when $\pi(d) \neq r$, M_r averages over the arcs obtained by applying each component-flip map in $\mathcal{T}_r(d)$ independently with probability $1/2$.

Lemma 6.2. *For every $r \in \mathcal{P}$, the operator M_r is self-adjoint with respect to $\langle \cdot, \cdot \rangle_\nu$ and satisfies $M_r^2 = M_r$.*

Proof. On each set of arcs generated by the component-flip maps, M_r is the product of the averages that apply each map with probability $1/2$. Each map permutes the arcs, preserves ν, and returns each arc to itself when applied twice, so its average is self-adjoint and idempotent. These averages commute, hence their product is self-adjoint and idempotent. The same properties hold on $\{d \in A \mid \pi(d) = r\}$, where $M_r = 0$. □

Suppose $|\{b \in r \mid b \in \pi(d)\}| = 1$ and $C \setminus \{e\} = D_1 \cup D_2$ has two nonempty components. Set $\Xi := b_r(\omega_c) \subseteq \Omega$. For each component $D \in K_r(\omega_c)$, use the local operators from Definition 5.3 with the abbreviations

$$R_D, A_D : \mathbb{R}^\Xi \longrightarrow \mathbb{R}^\Xi, \qquad R_D := R^\Xi_{r,\min D}, \qquad A_D := A^\Xi_{r,\min D}.$$

In particular, $(R_D\varphi)(\sigma) = \varphi(\Theta_{r,D}\sigma)$ and $A_D = (\mathrm{Id}_\Xi - R_D)/2$. The difference between the correlated and independent averages is the linear operator $W^\Xi_{r;D_1,D_2} : \mathbb{R}^\Xi \to \mathbb{R}^\Xi$ given by

$$\begin{aligned} W^\Xi_{r;D_1,D_2} &:= \left(\frac{\mathrm{Id}_\Xi + R_{D_1}R_{D_2}}{2} - \frac{(\mathrm{Id}_\Xi + R_{D_1})(\mathrm{Id}_\Xi + R_{D_2})}{4}\right) \prod_{D \neq D_1, D_2} \frac{\mathrm{Id}_\Xi + R_D}{2} \\ &= A_{D_1}A_{D_2} \prod_{D \neq D_1, D_2} \frac{\mathrm{Id}_\Xi + R_D}{2}. \end{aligned} \tag{6.4}$$

All products here range over $K_r(\omega_c)$. By Lemma 5.4, this operator is the orthogonal projection onto the one-dimensional subspace $\operatorname{span}\{\xi^\Xi_{\min D_1}\xi^\Xi_{\min D_2}\}$.

Define the linear correction map $D_r : \mathbb{R}^\Omega \to \mathbb{R}^A$ by

$$(D_r f)(d) := \begin{cases} (W^\Xi_{r;D_1,D_2}(f|_\Xi))(\omega_c), & d = (\omega_a, \omega_c, e),\ C \setminus \{e\} = D_1 \cup D_2, \\ -(W^\Xi_{r;D_1,D_2}(f|_\Xi))(\omega_c), & d = (\omega_c, \omega_a, e),\ C \setminus \{e\} = D_1 \cup D_2, \\ 0, & \text{otherwise}, \end{cases}$$

where the first two cases require $|\{b \in r \mid b \in \pi(d)\}| = 1$ and two nonempty components D_1, D_2 as above. Define $D := \sum_r D_r : \mathbb{R}^\Omega \to \mathbb{R}^A$.

Lemma 6.3. *For every $r \in \mathcal{P}$ and every $f : \Omega \to \mathbb{R}$, the operators defined above satisfy $\nabla P_r f = M_r \nabla f + D_r f$ in $\mathbb{R}^A$.*

Proof. Fix $d = (\omega, \tau, e) \in A$. If $r = \pi(d)$, then

$$P_r f(\omega) = P_r f(\tau), \qquad (M_r \nabla f)(d) = (D_r f)(d) = 0.$$

If $r \cap \pi(d) = \varnothing$, or if $|r \cap \pi(d)| = 1$ and $C \setminus \{e\}$ has at most one nonempty component, the two marginals of the coupled flips give

$$(M_r \nabla f)(d) = P_r f(\omega) - P_r f(\tau) = (\nabla P_r f)(d), \qquad (D_r f)(d) = 0.$$

It remains to consider $|r \cap \pi(d)| = 1$ and $C \setminus \{e\} = D_1 \cup D_2$ with both components nonempty. On $\Xi = b_r(\omega_c)$, the independent and correlated averages are

$$P_r f(\omega_c) = \left[\frac{(\mathrm{Id}_\Xi + R_{D_1})(\mathrm{Id}_\Xi + R_{D_2})}{4} \prod_{D \neq D_1, D_2} \frac{\mathrm{Id}_\Xi + R_D}{2}(f|_\Xi)\right](\omega_c),$$

$$(M_r \nabla f)(\omega_a, \omega_c, e) = P_r f(\omega_a) - \left[\frac{\mathrm{Id}_\Xi + R_{D_1}R_{D_2}}{2} \prod_{D \neq D_1, D_2} \frac{\mathrm{Id}_\Xi + R_D}{2}(f|_\Xi)\right](\omega_c).$$

Hence (6.4) gives

$$(\nabla P_r f - M_r \nabla f)(\omega_a, \omega_c, e) = (W^{\Xi}_{r;D_1,D_2}(f|_{\Xi}))(\omega_c),$$
$$(\nabla P_r f - M_r \nabla f)(\omega_c, \omega_a, e) = -(W^{\Xi}_{r;D_1,D_2}(f|_{\Xi}))(\omega_c).$$

These formulas agree with the definition of D_r and prove Lemma 6.3. □

Proof of Lemma 6.1. Use the operators constructed above. Lemma 6.2 gives the projection identities, and Lemma 6.3 gives the first equation in Lemma 6.1. Summing that equation over r and using $S = \sum_r (\mathrm{Id} - P_r)$ gives the second equation. □

6.2 Bounds for the coupled operators

Lemma 6.4. *There exists a universal constant $\mathsf{K}_{\mathrm{coup}} > 0$ such that, for every $qt^3 > \mathsf{K}_{\mathrm{coup}}$, every $f : \Omega \to \mathbb{R}$, and $\lambda \in (0, 1]$ satisfying $Sf = \lambda f$, the scalars $\mathsf{S}, \mathsf{T}, \mathsf{Y}$ defined above and $h := \kappa t^3 q$ satisfy*

$$\mathsf{Y} \geq \frac{h}{4}\mathsf{T}, \qquad \mathsf{Y}^2 \leq 2\Delta \mathsf{T}\mathsf{S}. \tag{6.5}$$

Proof of Theorem 3.2. The case $\mathsf{T} = 0$ is immediate. For $\mathsf{T} > 0$, choose K_2 large enough that $qt^3 \geq qt^{14} \geq \mathsf{K}_2$ satisfies the hypothesis of Lemma 6.4. With $h := \kappa t^3 q$, that lemma gives

$$\frac{h^2}{16}\mathsf{T}^2 \leq \mathsf{Y}^2 \leq 2\Delta \mathsf{T}\mathsf{S}, \qquad \mathsf{T} \leq \frac{32\Delta}{h^2}\mathsf{S} \leq \frac{32}{\kappa^2 q t^6}\mathsf{S}.$$

This proves the theorem with $\mathsf{C}_2 := 32/\kappa^2$. □

Lemma 6.5. *For every $f : \Omega \to \mathbb{R}$, with $g := \nabla f$, the constructed operators satisfy*

$$\left\langle g, \sum_r (\mathrm{Id} - M_r) g \right\rangle_{\nu} \geq \mathsf{T} + \mathsf{Y}.$$

Proof. For a fixed $e \in E$, define the finite set of exterior colorings

$$H_e := \{\omega|_{E\setminus\{e\}} \mid \omega \in \Omega\} \subseteq [q]^{E\setminus\{e\}}.$$

For $\eta \in H_e$, define its legal-color set and its cardinality by the maps

$$K_e : H_e \to 2^{[q]}, \qquad K_e(\eta) := \{c \in [q] \mid \eta(b) \neq c \text{ for every } b \in E \setminus \{e\} \text{ with } b \cap e \neq \varnothing\},$$
$$k_e : H_e \to \mathbb{Z}_{\geq 0},$$
$$k_e(\eta) := \big|\{c \in [q] \mid \eta(b) \neq c \text{ for every } b \in E \setminus \{e\} \text{ with } b \cap e \neq \varnothing\}\big| = |K_e(\eta)|.$$

For fixed η, abbreviate $K := K_e(\eta) \subseteq [q]$ and $k := k_e(\eta)$. For each $a \in K$, define $\omega_a : E \to [q]$ by $\omega_a(e) := a$ and $\omega_a|_{E\setminus\{e\}} := \eta$. Then $\omega_a \in \Omega$. The arcs with this exterior coloring are exactly (ω_a, ω_c, e) with $(a, c) \in K^2$ and $a \neq c$.

For every r, define the linear operator $L_r : \mathbb{R}^A \to \mathbb{R}^A$ by

$$(L_r\psi)(\omega_a, \omega_c, e) := \begin{cases} 0, & r = \{a, c\}, \\ \dfrac{1}{2}\displaystyle\sum_{b\in r} \psi(\omega_b, \omega_c, e), & a \in r,\ c \notin r,\ r \subseteq K, \\ \dfrac{1}{2}\displaystyle\sum_{b\in r} \psi(\omega_a, \omega_b, e), & c \in r,\ a \notin r,\ r \subseteq K, \\ \psi(\omega_a, \omega_c, e), & \text{otherwise.} \end{cases}$$

This definition is applied on every exterior-coloring class. The condition $r \subseteq K$ is equivalent to the absence of colors in r on edges adjacent to e. It is preserved by the simultaneous flip maps defining M_r. In either averaging case for L_r, e is a one-edge r-component and M_r is the product of L_r and the commuting projections for the other components. In the remaining cases L_r is zero or the identity. Hence, in the quadratic-form order for $\langle \cdot, \cdot \rangle_\nu$,

$$0 \preceq \mathrm{Id} - L_r \preceq \mathrm{Id} - M_r, \quad \text{that is, for every } \psi : A \to \mathbb{R}, \quad 0 \le \langle \psi, (\mathrm{Id} - L_r)\psi \rangle_\nu \le \langle \psi, (\mathrm{Id} - M_r)\psi \rangle_\nu.$$

Define the real scalars $u_a \coloneqq f(\omega_a)$ and $g_{ac} \coloneqq g(\omega_a, \omega_c, e) = u_a - u_c$. Direct substitution gives

$$\left[\sum_r (\mathrm{Id} - L_r) g\right](\omega_a, \omega_c, e) = (u_a - u_c) + \frac{1}{2} \sum_{b \in K \setminus \{a,c\}} [(u_a - u_b) + (u_b - u_c)] = \frac{k}{2} g_{ac}.$$

On arcs with $r \cap \{a, c\} = \varnothing$, one has $L_r = \mathrm{Id}$, so this sum does not include their contribution from $\mathrm{Id} - M_r$. Summing that contribution separately gives $\mathcal{B}_T(f)$ in (6.1): for $d = (\omega, \tau, e)$ with $p \coloneqq \pi(d)$, $g(d) = 2G_{e,p} f(\omega)$ and $\nu(\{d\}) = \mu(\omega)/4$. For each r, the three subsets of A specified by $\pi(d) = r$, $|\{b \in r \mid b \in \pi(d)\}| = 1$, and $\pi(d) \cap r = \varnothing$ are invariant under every flip map defining M_r. The corresponding quadratic forms therefore add. Since $k = H(d) + 2$ on every arc in the class,

$$\left\langle g, \sum_r (\mathrm{Id} - M_r) g \right\rangle_\nu \ge \int_A (1 + H/2) g^2 \, d\nu + \mathcal{B}_T(f) = \mathsf{T} + \mathsf{Y}.$$

□

Lemma 6.6. *For every $f : \Omega \to \mathbb{R}$, the correction maps satisfy*

$$\sum_r \|D_r f\|_\nu^2 \le \mathsf{S}, \qquad \|Df\|_\nu^2 \le 2\Delta\mathsf{S}.$$

Proof. Fix $r \in \mathcal{P}$ and use the family B_r of conditioning classes defined in Section 5.1. For $\Xi \in B_r$, set $s_\Xi \coloneqq |K_r(\sigma)|$ for any $\sigma \in \Xi$, and enumerate its component edge sets as $D_1, \dots, D_{s_\Xi}$. Define the endpoint-set map $\partial : 2^E \to 2^V$ by

$$\partial D \coloneqq \{v \in V \mid |\{b \in D \mid v \in b\}| = 1\}.$$

In particular, ∂D has two vertices for a nonempty path component and is empty for a cycle component. Define the index set and the function

$$\begin{aligned} J_r &\coloneqq \{(\Xi, i, j) \mid \Xi \in B_r,\ 1 \le i < j \le s_\Xi\}, \\ n_r &: J_r \longrightarrow \{0, 1, 2, 3, 4\}, \\ n_{ij}(\Xi) &\coloneqq n_r(\Xi, i, j) \coloneqq |\{\{u, v\} \in E \mid u \in \partial D_i,\ v \in \partial D_j\}|. \end{aligned}$$

The bound $n_{ij}(\Xi) \le 4$ follows because G is simple. For each $(\Xi, i, j) \in J_r$, define the global linear operator $W_{\Xi,ij} : \mathbb{R}^\Omega \to \mathbb{R}^\Omega$ as the extension by zero of $W^\Xi_{r;D_i,D_j}$:

$$(W_{\Xi,ij} f)(\sigma) \coloneqq \begin{cases} (W^\Xi_{r;D_i,D_j}(f|_\Xi))(\sigma), & \sigma \in \Xi, \\ 0, & \sigma \notin \Xi. \end{cases}$$

For path components D_i, D_j, each edge joining their endpoints contributes $\frac{1}{4}\|W_{\Xi,ij} f\|_\mu^2$ to $\|D_r f\|_\nu^2$. Indeed, exactly two of the four joint orientations of D_i, D_j have equal colors beside that edge.

In each of these two orientations precisely one color in r can be assigned to the edge, and both orientations of the resulting arc have ν-mass $1/(4\mathsf{Z})$. By (6.4), the function $W_{\Xi,ij}f$ is a scalar multiple of a product basis function on Ξ, so its square is constant there. The operators $W_{\Xi,ij}$ for distinct triples have orthogonal ranges, all contained in the range of Q_r. Consequently

$$\|D_r f\|_\nu^2 = \sum_{\Xi\in B_r}\sum_{i<j}\frac{n_{ij}(\Xi)}{4}\|W_{\Xi,ij}f\|_\mu^2 \le \|Q_r f\|_\mu^2. \tag{6.6}$$

Summing over r and using $\mathsf{S}=\sum_r\|Q_rf\|_\mu^2$ proves the first asserted bound.

For an arc $d=(\omega,\tau,e)$ with $e=\{u,v\}$, $a:=\omega(e)$, and $c:=\tau(e)$, define the finite set

$$J(d):=\{\gamma\in[q]\mid \exists b_u,b_v\in E\setminus\{e\},\ u\in b_u,\ v\in b_v,\ \omega(b_u)=\omega(b_v)=\gamma\}.$$

This specifies a map $J:A\to 2^{[q]}$ and $|J(d)|\le\Delta$. If $(D_rf)(d)\neq 0$, then $r\in\{\{a,\gamma\},\{c,\gamma\}\mid\gamma\in J(d)\}$. There are at most 2Δ such pairs. Applying Cauchy–Schwarz pointwise and integrating gives

$$\|Df\|_\nu^2\le 2\Delta\sum_r\|D_rf\|_\nu^2\le 2\Delta\mathsf{S}. \tag{6.7}$$

□

Lemma 6.7. *There exists a positive universal constant K_{Y} such that, for every $qt^3>\mathsf{K}_{\mathrm{Y}}$ and every $f:\Omega\to\mathbb{R}$, we have $\mathsf{Y}\ge h\mathsf{T}/4$, where $h:=\kappa t^3q$.*

Proof. Set $\rho:=\mathsf{A}e^{-\mathsf{c}qt^3}$, using the constants from Theorem 4.1. Apply that theorem to $G_{e,p}f:\Omega\to\mathbb{R}$ for every e,p, and sum. Since $g(\omega,\tau,e)=2G_{e,p}f(\omega)$ for $p:=\{\omega(e),\tau(e)\}$, the real scalar

$$\mathsf{T}_{\le h}:=\int_A g^2\mathbf{1}_{\{H\le h\}}\,d\nu\quad\text{satisfies}\quad \mathsf{T}_{\le h}\le\frac{\mathsf{A}}{q^2t^6}\mathcal{B}_T(f)+\rho\mathsf{T}. \tag{6.8}$$

The definition of Y and (6.8) give

$$\begin{aligned}\mathsf{Y}&\ge\frac{h}{2}(\mathsf{T}-\mathsf{T}_{\le h})+\mathcal{B}_T(f)\\&\ge\frac{h}{2}(1-\rho)\mathsf{T}+\left(1-\frac{\mathsf{A}h}{2q^2t^6}\right)\mathcal{B}_T(f)\ge\frac{h}{4}\mathsf{T}.\end{aligned}$$

The last inequality holds for sufficiently large qt^3, because $\rho=\mathsf{A}e^{-\mathsf{c}qt^3}\le 1/2$ and $\mathsf{A}h/(2q^2t^6)=\mathsf{A}\kappa/(2qt^3)\le 1$. □

Proof of Lemma 6.4. Lemma 6.7 gives $\mathsf{Y}\ge h\mathsf{T}/4$. Lemmas 6.5 and 6.6 give

$$\left\langle g,\sum_r(\mathrm{Id}-M_r)g\right\rangle_\nu\ge\mathsf{T}+\mathsf{Y},\qquad\sum_r\|D_rf\|_\nu^2\le\mathsf{S}. \tag{6.9}$$

The eigenfunction identity $Sf=\lambda f$ implies $\nabla Sf=\lambda g$. Taking the inner product with g in the second identity of Lemma 6.1 and using Lemma 6.5 gives

$$\langle g,Df\rangle_\nu=\left\langle g,\sum_r(\mathrm{Id}-M_r)g\right\rangle_\nu-\lambda\mathsf{T}\ge\mathsf{Y}+(1-\lambda)\mathsf{T}\ge\mathsf{Y}.$$

Cauchy–Schwarz and Lemma 6.6 now yield $\mathsf{Y}^2\le\mathsf{T}\|Df\|_\nu^2\le 2\Delta\mathsf{TS}$. □

A Component operators and decomposition

This appendix constructs the operators in Lemma 5.1 and proves their properties used in Sections 5 and 6.

A.1 Construction of the component operators

Fix a total order on E. For each color pair $p \in \mathcal{P}$ and edge label $i \in E$, define the map $C_{p,i} : \Omega \to 2^E$ by

$$C_{p,i}(\sigma) := \begin{cases} C, & C \in K_p(\sigma) \text{ and } \min C = i, \\ \varnothing, & \text{if no such component exists.} \end{cases}$$

We call $C_{p,i}$ the *indexed Kempe component map*. For each fixed coloring σ and pair p, every nonempty Kempe component $C \in K_p(\sigma)$ has exactly one label $i := \min C$. Thus the pair (p, i) specifies its color pair and its unique label among the components of that pair, with each p-component included exactly once.

Recall the coloring-class map $b_p : \Omega \to 2^\Omega$ from Definition 1.1, and define its family of distinct classes by

$$B_p := \{b_p(\sigma) \mid \sigma \in \Omega\} \subseteq 2^\Omega.$$

These classes partition Ω. Each $C_{p,i}$ is constant on every $\Xi \in B_p$. For fixed p, Ξ, define the component edge sets and their index set by

$$C_i := C_{p,i}(\sigma) \subseteq E \quad (\sigma \in \Xi), \qquad I_{p,\Xi} := \{i \in E \mid C_i \neq \varnothing\} \subseteq E.$$

In particular, Ξ is a set of colorings, whereas C_i is an edge set of one component. The maps $M_{p,e}$ and $H_{p,e}$ are also constant on each class. We write $M_{p,e}(\Xi)$ and $H_{p,e}(\Xi)$ for their common values there.

For every $\Xi \in B_p$, the flip Θ_{p,C_i} preserves Ξ and is the identity when $i \notin I_{p,\Xi}$.

Use the local operators from Definition 5.3. If $i \notin I_{p,\Xi}$, then $R_{p,i}^\Xi = \mathrm{Id}_\Xi$ and $A_{p,i}^\Xi = F_{p,i}^\Xi = 0$.

The operators on the full state space have separate notation:

$$R_{p,i},\ A_{p,i},\ N_p,\ F_{p,i}.$$

For every $g : \Omega \to \mathbb{R}$ and $\Xi \in B_p$, define them by the restriction formulas

$$\begin{aligned} (R_{p,i}g)|_\Xi &:= R_{p,i}^\Xi(g|_\Xi), \quad (A_{p,i}g)|_\Xi := A_{p,i}^\Xi(g|_\Xi), \\ (N_p g)|_\Xi &:= N_p^\Xi(g|_\Xi), \quad (F_{p,i}g)|_\Xi := F_{p,i}^\Xi(g|_\Xi). \end{aligned}$$

These formulas determine the operators uniquely because the classes partition Ω. Each operator acts separately on the coloring classes. For example,

$$F_{p,i} = \bigoplus_{\Xi \in B_p} F_{p,i}^\Xi, \qquad \|F_{p,i}g\|_\mu^2 = \sum_{\Xi \in B_p} \|F_{p,i}^\Xi(g|_\Xi)\|_{\mu,\Xi}^2.$$

Thus a superscript Ξ denotes the restriction to that class. Omitting it denotes the operator on all of Ω. Recall that $\mathsf{Z} := |\Omega|$ is the number of proper colorings. It is a scalar, while $\mathbb{Z}$ denotes the set of integers. Restricted norms follow the normalization convention in Section 2.1.

A.2 Proof of the component-basis lemma

Proof of Lemma 5.4. Fix $p \in \mathcal{P}$ and $\Xi \in B_p$, and retain the notation of Definition 5.2, including $I := I_{p,\Xi}$ and $k := |I|$. Each nonempty two-color component has exactly two proper colorings, which differ by exchanging the two colors of p. The colors outside p are fixed on Ξ and impose no additional condition on these choices. Choices on distinct components are independent in the following precise sense: the coordinate map

$$\Xi \longrightarrow \{-1,1\}^I, \qquad \sigma \longmapsto (\xi_i^\Xi(\sigma))_{i\in I},$$

is a bijection. Indeed, a sign vector specifies whether each component has its reference coloring or its flipped coloring, which specifies exactly one coloring in Ξ. Consequently, $|\Xi| = 2^k$.

For subsets $J, K \subseteq I$, define their symmetric difference $J\triangle K := (J \setminus K) \cup (K \setminus J) \subseteq I$. Since $(\xi_i^\Xi)^2 = 1$ pointwise,

$$\psi_J^\Xi \psi_K^\Xi = \psi_{J\triangle K}^\Xi.$$

If $J = K$, this product is identically one. If $J \neq K$, choose $i \in J\triangle K$. The bijection of $\{-1,1\}^I$ that changes only coordinate i pairs every term in the sum of $\psi_{J\triangle K}^\Xi$ with its negative. Therefore

$$\frac{1}{\mathsf{Z}} \sum_{\sigma\in\Xi} \psi_J^\Xi(\sigma)\psi_K^\Xi(\sigma) = \frac{|\Xi|}{\mathsf{Z}} \mathbf{1}_{\{J=K\}}.$$

Each basis candidate has nonzero squared norm $|\Xi|/\mathsf{Z}$, and their number is $2^k = \dim \mathbb{R}^\Xi$. They therefore form an orthogonal basis. With the normalized uniform probability measure on Ξ, the corresponding squared norm is one. The restricted norm in this paper instead assigns mass $1/\mathsf{Z}$ to each coloring.

For $i \in I$, flipping component i replaces ξ_i^Ξ by $-\xi_i^\Xi$ and preserves every other coordinate. Thus

$$R_{p,i}^\Xi \psi_J^\Xi = (-1)^{\mathbf{1}_{\{i\in J\}}} \psi_J^\Xi, \qquad A_{p,i}^\Xi \psi_J^\Xi = \frac{1-(-1)^{\mathbf{1}_{\{i\in J\}}}}{2} \psi_J^\Xi = \mathbf{1}_{\{i\in J\}} \psi_J^\Xi.$$

These formulas also hold for $i \notin I$, since then $R_{p,i}^\Xi = \mathrm{Id}_\Xi$ and $A_{p,i}^\Xi = 0$. Summing the second identity over $i \in E$ gives

$$N_p^\Xi \psi_J^\Xi = |J| \psi_J^\Xi.$$

In particular, N_p^Ξ is self-adjoint and positive semidefinite, and its kernel consists precisely of the constant functions on Ξ.

The linear map $(N_p^\Xi)^{-1/2} : \mathbb{R}^\Xi \to \mathbb{R}^\Xi$ assigns multiplier $a^{-1/2}$ to an eigenvector of eigenvalue $a > 0$ and multiplier zero to an eigenvector of eigenvalue zero. The eigenvalue formula above therefore gives

$$(N_p^\Xi)^{-1/2} \psi_J^\Xi = \begin{cases} |J|^{-1/2} \psi_J^\Xi, & J \neq \varnothing, \\ 0, & J = \varnothing. \end{cases}$$

Since $F_{p,i}^\Xi := A_{p,i}^\Xi (N_p^\Xi)^{-1/2}$, this proves the claimed formula for $F_{p,i}^\Xi$. All four local operators are diagonal in the orthogonal basis just constructed and hence are self-adjoint.

For $J \neq \varnothing$, the sum of the squared multipliers of $F_{p,i}^\Xi$ is

$$\sum_{i\in E} \frac{\mathbf{1}_{\{i\in J\}}}{|J|} = 1.$$

For $J = \varnothing$, every multiplier is zero. The averaging operator P_p^{Ξ} fixes the constant basis function and annihilates every other basis function, by the proved orthogonality. Consequently $Q_p^{\Xi} \coloneqq \mathrm{Id}_{\Xi} - P_p^{\Xi}$ has exactly these multipliers, and

$$\sum_{i \in E} (F_{p,i}^{\Xi})^2 = Q_p^{\Xi}.$$

Finally, for $J \neq \varnothing$, the multipliers satisfy

$$0 \leq \frac{\mathbf{1}_{\{i \in J\}}}{\sqrt{|J|}} \leq \mathbf{1}_{\{i \in J\}} \leq 1,$$

for $J = \varnothing$, the first two multipliers are zero. Expanding an arbitrary map $g : \Xi \to \mathbb{R}$ in the orthogonal basis and comparing the squared coefficients proves

$$\|F_{p,i}^{\Xi} g\|_{\mu,\Xi} \leq \|A_{p,i}^{\Xi} g\|_{\mu,\Xi} \leq \|g\|_{\mu,\Xi}.$$

□

A.3 Proof of the decomposition lemma

Proof of Lemma 5.1. Applying Lemma 5.4 on every coloring class and summing with mass $1/\mathsf{Z}$ per coloring gives

$$\begin{aligned} \sum_{i \in E} F_{p,i}^2 &= Q_p \quad (p \in \mathcal{P}), \\ \sum_{p \in \mathcal{P}} \sum_{i \in E} \|F_{p,i} g\|_\mu^2 &= \mathcal{E}_S(g), \qquad \|F_{p,i} g\|_\mu \leq \|A_{p,i} g\|_\mu \leq \|g\|_\mu, \end{aligned} \tag{A.1}$$

for every function $g : \Omega \to \mathbb{R}$. For the eigenfunction f fixed at the beginning of Section 5, the second identity reads

$$\sum_{p \in \mathcal{P}} \sum_{i \in E} \|F_{p,i} f\|_\mu^2 = \mathcal{E}_S(f) = \mathsf{S}.$$

The applications of Theorem 4.1 to $F_{p,i} f$ involve $\mathcal{E}_{\mathrm{aux},p}(F_{p,i} f)$. An update with $r \cap p = \varnothing$ preserves E_p and commutes with every $F_{p,i}$. Hence Lemma 5.4 gives

$$\sum_{p,i} \sum_{r \cap p = \varnothing} \|Q_r F_{p,i} f\|_\mu^2 = \sum_{p \cap r = \varnothing} \|Q_r Q_p f\|_\mu^2. \tag{A.2}$$

The sum on the right is over ordered pairs (p, r) of disjoint color pairs. For $a \in [q]$, define the linear operator L_a on $\mathbb{R}^\Omega$ by $L_a \coloneqq \sum_{p \ni a} Q_p$. Expanding squares gives

$$\sum_a L_a^2 = S^2 + S - \sum_{p \cap r = \varnothing} Q_p Q_r.$$

Since $0 \preceq \sum_a L_a^2$ and the disjoint products $Q_p Q_r$ are orthogonal projections,

$$\sum_{p \cap r = \varnothing} \|Q_r Q_p f\|_\mu^2 \leq \|Sf\|_\mu^2 + \mathsf{S} \leq 2\mathsf{S} \qquad \text{if } Sf = \lambda f, \quad 0 < \lambda \leq 1. \tag{A.3}$$

□

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